

THE LIMITING MOBILITIES OF IONS

THE transport of electricity through electrolytes differs fundamentally from metallic conduction in that the carriers are ions, the dimensions and masses of which are much larger than those of the electrons responsible for metallic conduction. The ions of course share in the general Brownian motion of the liquid, and may be expected to have randomly-directed instantaneous velocities of the order of 10^4 cm sec⁻¹, though of course with the extremely short mean free path characteristic of the liquid state. In the absence of an external field or a concentration-gradient, the Brownian movement is entirely random, and does not lead to a drift of ions in any one direction. The presence of an electric field, as in conductance, or of a concentration-gradient, as in diffusion, has the effect of biasing the Brownian movement in a particular direction. In a field of 1 V/cm the average velocity of the ions in the direction of the field is of the order of 10^{-3} to 10^{-4} cm sec⁻¹, and hence represents only a very small perturbation of the random ionic motions. The actual path of an ion under an electric field of ordinary intensity is thus extremely erratic, bearing very little resemblance to that of a billiard-ball sinking in water. Nevertheless, the drastic simplification of substituting for the actual chaotic motion a steady progress of all the ions of one kind with equal velocities in one direction of the field is extraordinarily successful: the Brownian motion needs to be considered only in regard to its effect on the interionic forces.

Experimental data on conductivity are fortunately extremely plentiful and of high accuracy, at least for low concentrations; in the best work agreement to one part in 10,000 between different workers is not uncommon. Especially in non-aqueous solutions and in mixed solvents, conductivity measurements are far more easily made than those of activities, and provide the greater part of our knowledge of the behaviour of electrolytes in such solutions. Furthermore, the measurements can be carried to extraordinarily low concentrations provided proper precautions are taken. Whereas the measurement of the electromotive forces of cells usually becomes unreliable at concentrations below about 0.001 M even in the most favourable cases, accurate conductivity measurements can be made at concentrations down to about 0.00003 M. The experimental

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techniques have been discussed in Chapter 5; here we are concerned with the theoretical interpretation of the results.

The equivalent conductivity of strong electrolytes at low concentrations is found to be accurately a linear function of the square root of concentration, decreasing as the concentration increases. Extrapolation to zero concentration yields the limiting equivalent conductivity Λ^0 , and the equivalent conductivity Λ , at *very low concentrations* can therefore be represented by the equation:

$$\Lambda = \Lambda^0 - A\sqrt{c} \quad \dots (6.1)$$

as was observed by Kohlrausch.

A complete theory of electrolytic conduction should therefore be capable of (a) predicting the value of Λ^0 from the dimensions, charges and other properties of the ions and the solvent molecules, (b) predicting the value of the constant A in equation (6.1), (c) accounting quantitatively for deviations from equation (6.1) at higher concentrations. Of these three problems, the first is farthest from solution, the second is solved and the position with regard to the third has recently been greatly improved.

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In the state of infinite dilution to which Λ^0 refers, the motion of an ion is limited solely by its interactions with the surrounding solvent molecules, there being no other ions within a finite distance. In these circumstances, the validity of Kohlrausch's law of the independent migration of ions is almost axiomatic; according to this law each species of ion present contributes at infinite dilution a definite amount to the total equivalent conductivity, regardless of the nature of the other ions present. Thus for an electrolyte giving two kinds of ions,

$$\Lambda^0 = \lambda_1^0 + \lambda_2^0 \quad \dots (6.2)$$

The values of λ_1^0 and λ_2^0 may be determined by measurements of transport numbers t , which may also be extrapolated linearly to infinite dilution against the square root of concentration. Thus

$$\begin{aligned} \lambda_1^0 &= t_1^0 \Lambda^0 \\ \lambda_2^0 &= t_2^0 \Lambda^0 \end{aligned} \quad \dots (6.3)$$

The accuracy with which such measurements confirm Kohlrausch's law of the independent migration of ions may be seen from the data in Table 6.1 for aqueous potassium and sodium chloride compiled from papers by GORDON and his collaborators⁽¹⁾. These measurements represent probably the most accurate test yet made of the Kohlrausch principle; it will be noted that even at 45° where

the experimental difficulties are most marked, the two independent values of λ_{Cl}^0 agree within 0.04 per cent. Thus the step of resolving Λ^0 for a salt into values of λ^0 for its separate ions can be taken with complete confidence provided that accurate values of the transport numbers are available at concentrations low enough to permit extrapolation to zero concentration. To account for the observed

Table 6.1
Test of Kohlrausch's Law of the Independent Migration of Ions

Temp. °C	15°	25°	35°	45°
Λ^0 KCl	121.07	149.85	180.42	212.41
λ_{Cl}^0 (KCl)	0.5072	0.5095	0.5111	0.5128
λ_{K}^0 (KCl)	61.41	76.35	92.21	108.92
Λ^0 NaCl	101.18	126.45	153.75	182.65
λ_{Cl}^0 (NaCl)	0.6071	0.6038	0.5998	0.5961
λ_{Na}^0 (NaCl)	61.43	76.35	92.22	108.88

Λ^0 in (cm² Int. Ω^{-1} equiv.⁻¹)

values of λ^0 in terms of other properties of the ions is, however, a much more difficult problem, of which at present only a qualitative treatment can be given.

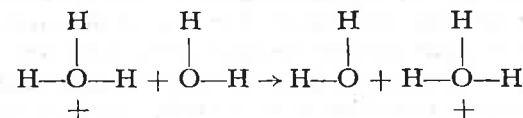
The limiting equivalent conductivities of a number of ions at 25° in water are compiled in Appendix 6.1. These have been obtained as follows: the best available data for Λ^0 for various salts have been selected from the literature as indicated by the references quoted. In the case of chlorides, the cation mobility λ_{Cl}^0 has been computed as $\lambda_{Cl}^0 = \Lambda^0 - \lambda_{Cl}^0$ using the value $\lambda_{Cl}^0 = 76.35$ obtained from Gordon's data (Table 6.1). The values for other anions have then been computed as $\lambda_{Cl}^0 = \Lambda^0 - \lambda_{Cl}^0$ using wherever possible Λ^0 values for potassium or sodium salts and the tabulated values for λ_{Na}^0 or λ_{K}^0 . This is done to ensure the self-consistency of the tabulated values, but means that in some cases the value given for λ^0 is not quite that decided upon by the workers referred to, owing to a different choice of the limiting transport numbers. The table will, however, permit the calculation of Λ^0 from the constituent λ^0 values within the experimental error.

THE INTERPRETATION OF THE LIMITING EQUIVALENT CONDUCTIVITIES OF IONS

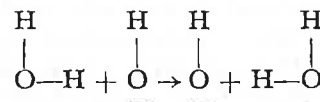
The most striking feature of the ionic conductivities compiled in Appendix 6.1 is the extremely high mobility of the hydrogen ion, which clearly suggests that a special mechanism is involved in its motion. It is scarcely possible to imagine that the bare proton could be moving freely through the solution, for this would lead to

an almost infinite mobility. Nor is it possible to regard the moving entity as the H_3O^+ ion, (though this formula is often written for the aqueous hydrogen ion) since this ion would have dimensions similar to those of a water molecule, and the mobility of the water molecule is known from experiments on the self-diffusion of water to be similar to that of simple ions such as K^+ and Cl^- .† A reasonable explanation has been found⁽²⁾ in terms of a 'proton jump' mechanism, by which a proton passes from one water molecule to a favourably oriented neighbouring one, but in doing so leaves these molecules unfavourably oriented for another jump.

At any one time, only a few of the protons in a solution will be indulging in these 'jumps'. The majority will be definitely associated with one water molecule or another, and to this extent it is legitimate to write the hydrogen ion as H_3O^+ . However, it is believed that this ion can fit into the normal coordinated structure of water almost as well as can an ordinary water molecule, so the charged molecule may become the centre of a rather firmly associated group of water molecules; it may, in fact, become further hydrated. This would explain the remarkable similarity between the activity coefficients of lithium chloride, bromide, iodide and perchlorate and those of the corresponding acids, which implies that from a thermodynamic point of view the hydrated lithium ion and the hydrogen ion are of nearly the same size and involve about the same number of water molecules, while the proton-jump mechanism accounts for the fact that the mobility of the hydrogen ion under an applied electrical field is some ten times that of the lithium ion. The suggested proton jump mechanism can be represented diagrammatically (after GLASSTONE, LAIDLER and EYRING⁽²⁾) as follows:



The abnormally high mobility of the aqueous hydroxide ion, which is second only to that of hydrogen ion, may be similarly accounted for by the proton-transfer process:



† The mobility of the H_3O^+ ion as a unit may be approximately calculated in units of equivalent conductivity as $D^* = RT\lambda/(|z|F^2)$ where D^* is the self-diffusion coefficient of water ($\sim 2.4 \times 10^{-5}$ cm² sec⁻¹ at 25°). This gives $\lambda_{H_3O^+}^0 \approx 90$ cm² Ω^{-1} equiv.⁻¹.

If $(\Lambda_{\text{HCl}}^0 - \Lambda_{\text{KCl}}^0)/\Lambda_{\text{KCl}}^0$ is taken as a measure of the abnormal hydrogen ion mobility, the data in Appendix 6.2 give 2.26, 1.84 and 1.07 for this ratio at 0°, 25° and 100° respectively, suggesting that the breaking of the water structure reduces the abnormal mobility. A pressure of 3,000 atm.,^(2a) however, increases the abnormal mobility, the ratio being 2.15 compared with 1.84 at 1 atm.

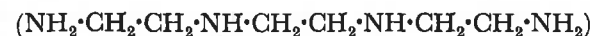
The equivalent conductivity of hydrochloric acid which is 426.1 in water at 25° is only 198.5 cm² Ω⁻¹ equiv⁻¹ in methanol whilst it has a minimum value in a water-methanol mixture containing about 10 per cent by weight of water⁽³⁾. In this mixed solvent its conductivity is similar to that of sodium chloride. It is evident that the proton jump via the CH₃OH₂⁺ complex is less effective than it is via the H₃O⁺ complex and that the abnormal mobility is absent in 90 per cent methanol.

Having eliminated these two exceptional cases, we find that some further interesting generalizations emerge from an inspection of Appendix 6.1. The maximum mobility of monovalent ions (at 25° in water) is about 75 equivalent conductivity units; the mobilities of K⁺, Tl⁺, NH₄⁺, Cl⁻, Br⁻, I⁻, NO₃⁻, ClO₄⁻ all cluster closely about this value. It appears that these ions lie in a critical range of size: if they were smaller (in terms of crystallographic radius) they would acquire a permanent hydration sheath and end up larger and with lower mobility, as do sodium, lithium and fluoride ions; if they were larger in crystallographic radius, they would not hydrate, but would be slower-moving merely on account of their size like, for example, the carboxylic acid anions.

The order of the mobilities of the alkali-metal cations is the inverse order of their crystallographic size, which is of course in accordance with the expectation that ions of the greatest surface charge will be most strongly hydrated. The same order holds for the bivalent cations, though the practically identical values for Ca⁺⁺ and Sr⁺⁺ suggest that these two hydrated ions have very similar dimensions. (This similarity is not so marked in the activity coefficients of calcium and strontium salts.) The mobilities of the bivalent cations cover only a small range, about 53–63 units; this may well be because they all have one firmly attached layer of water molecules and only a few in a second layer. Among the few bivalent anions for which data are available the symmetrical tetrahedral sulphate ion shows a substantially higher mobility than the others and even than the bivalent cations, suggesting that it is sufficiently 'padded' with oxygen atoms to prevent any extensive hydration. Comparison of the structurally rather similar sulphate and perchlorate ions on the basis of Stokes' law radius,

$r = 0.820|z|/(\lambda^0\eta^0)$, however, indicates that the sulphate ion has a substantially larger 'radius', about 70 per cent greater than that of ClO₄⁻.

The trivalent cations of the rare earths, as might be expected, all show very similar mobilities, close to $\lambda^0 = 70$; the ions are evidently all hydrated to much the same large extent. This conclusion is confirmed by the fact that the activity coefficient data for their chlorides all require values of the ion size parameter a which lie in the range 5.6–6.0 Å. Their mobilities are strikingly lower than those of the trivalent complex ions Co(NH₃)₆⁺⁺⁺ and Fe(CN)₆⁻⁻⁻, which are both close to 100 units; in these ions the place of the first layer of water molecules is taken by NH₃ and CN⁻ respectively, and water molecules do not appear to attach themselves to these 'foreign' groups as readily as to other water molecules. The various polyphosphate ions, which have been carefully investigated by DAVIES and MONK⁽⁴⁾, provide interesting examples of anions with high negative charges, and attention should also be drawn to JAMES' study⁽⁵⁾ of the sexavalent cation [Co₂ trien₃]⁶⁺ of *tris*-triethylenetetraminecobaltic chloride, a quadridentate compound containing two cobalt atoms and three triethylenetetramine



molecules.

The tetra-alkyl ammonium ions⁽⁶⁾ are of great theoretical interest because they combine large size and symmetrical shape with low charge, and furthermore, some of their salts are soluble in many solvents besides water. In the other solvents, however, the individual ionic mobilities are less certainly known because (a) the conductivity measurements are in general less easily extrapolated to infinite dilution and (b) the limiting transport numbers are seldom known experimentally but have to be guessed on some reasonable basis. The λ^0 values used in compiling Appendix 6.1 for these ions in water at 25° represent the latest values given by KRAUS and his collaborators⁽⁷⁾, obtained from measurements by the most fastidious techniques extending to concentrations as low as 10⁻⁴ molar, and are almost certainly to be preferred to the numerous earlier values to be found in the literature. They are of great value as a test of the validity of Stokes' law for ions in aqueous solutions. There is strong reason to believe, from an examination of the temperature-dependence of ionic mobility (see pp. 128–129) that for ions which are (a) intrinsically large and of low surface charge, or (b) of sufficiently large surface charge to form firmly hydrated entities, Stokes' law is of the correct form though the numerical constant

may not be 6π . For these ions the product $\lambda^0\eta^0$ is very nearly constant over a fair range of temperatures in water. The possibility therefore presents itself that we might use the mobilities of the tetra-substituted ammonium ions to calculate correction factors for Stokes' law in water, and then by using these factors, calculate the size of the strongly hydrated ions from their mobilities. In order to do this it is of course necessary to know the sizes of the substituted ammonium ions. A fair approximation to their sizes may be obtained as follows:

1. The effective radius of the $\text{N}(\text{CH}_3)_4^+$ ion can be estimated from the N-C internuclear distance of 1.47 Å to which is added Pauling's value of 2.0 Å for the Van der Waals radius of the methyl group as a whole, giving 3.47 Å.

2. For the ion $\text{N}(\text{C}_2\text{H}_5)_4^+$ a similar calculation from bond-lengths and angles indicates a maximum radius of about 4.2 Å, while a scale model (using 'Catalin' atomic models) suggests an average radius of about 4.0 Å; the value is somewhat dependent on the configuration given to the C-C-H linkages. The latter value is probably preferable.

3. For the higher homologues, it is not easy to estimate a radius from bond lengths or models, as too many configurations exist. The following rather tentative method may be tried: the first two members of the series are structurally very similar to the symmetrical paraffins $\text{C}(\text{CH}_3)_4$ and $\text{C}(\text{C}_2\text{H}_5)_4$ which have molal volumes of approximately 120 cm³ and 170 cm³ respectively. One would expect the radii to be directly proportional to the cube roots of the molecular volumes, and one finds, in fact, that the empirical relation:

$$r \approx 0.72 \bar{V}^{1/3}$$

(with r in Å and $\bar{V}$ in cm³ per mole) gives for the first two members $r = 3.55$ Å and $r = 3.99$ Å in adequate agreement with the values given above. One may then estimate approximate radii for the higher members by this formula, assuming for the density of the corresponding paraffins the value of 0.75 which is typical of the higher paraffins. The radii of the ions calculated in this way are given in the column headed r in Table 6.2. The Stokes' law radii obtained from the limiting mobilities of Appendix 6.1 are given in the column headed r_s ; since the viscosity of water at 25° is 0.008903 poise, equation 2.49 becomes $r_s = 92.1/\lambda^0$ for monovalent ions. The ratio r/r_s can be regarded as a correction factor for Stokes' law in water and the table suggests that the law is applicable for particles greater than ~ 5 Å in radius, but gives radii which are

Table 6.2

Ion	r (Å)	r_s (Å)	r/r_s
$\text{N}(\text{CH}_3)_4^+$	3.47	2.05	1.69
$\text{N}(\text{C}_2\text{H}_5)_4^+$	4.00	2.82	1.42
$\text{N}(\text{C}_3\text{H}_7)_4^+$	4.52	3.93	1.15
$\text{N}(\text{C}_4\text{H}_9)_4^+$	4.94	4.73	1.04
$\text{N}(\text{C}_6\text{H}_{11})_4^+$	5.29	5.27	1.00

r = radius estimated from molecular volumes or models.

r_s = radius calculated from the limiting mobility by Stokes' law.

considerably too small when applied to particles smaller than this. The correction factor is plotted against the Stokes' law radius in Figure 6.1. We may now very tentatively use this graph to estimate

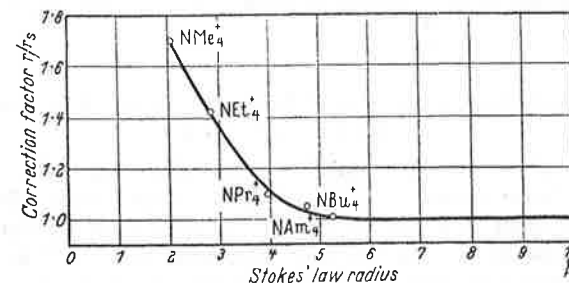


Figure 6.1. Tentative correction factors for Stokes' law in water at 25°

the radii of heavily hydrated ions from their limiting mobilities, assuming these corrections to apply. The results for a number of ions are given in Table 6.3; the calculation is of course confined to cases where the ion is of symmetrical shape and has a Stokes' law radius, $r_s = 0.820|z|/(\lambda^0\eta^0)$, in the range above 2.0 Å.

The 'corrected Stokes' law radius' of the hydrated ion can then be used to estimate its volume, and since the volume of the bare ion itself is negligible compared to the resulting values, a rough estimate can then be made of the average number of water molecules involved in the hydrated entity by neglecting the electrostriction of these molecules, and ascribing to them their ordinary liquid volume of 30 Å³. The hydration numbers 'h' so obtained are given in the last column of Table 6.3, and it must be admitted that they are eminently reasonable.

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Table 6.3

Estimates of the Radii of Hydrated Ions from Modified Stokes' Law:

$$r = \frac{0.820|z|}{\lambda^0 \eta^0} \left(\frac{r}{r_s} \right) \text{ in } \text{\AA}$$

Ion	λ^0	r_s	r	r (crystal- lographic)	$\frac{4}{3}\pi r^3$ (cu \AA)	h
Na ⁺	50.10	1.83	3.3	0.97	150	5
Li ⁺	38.68	2.37	3.7	0.60	210	7
Be ⁺⁺	45	4.08	4.6	—	410	13-14
Mg ⁺⁺	53.05	3.46	4.4	0.65	360	12
Ca ⁺⁺	59.50	3.09	4.2	0.99	310	10
Sr ⁺⁺	59.45	3.09	4.2	1.13	310	10
Ba ⁺⁺	63.63	2.88	4.1	1.35	290	9-10
Zn ⁺⁺	53.0	3.46	4.4	0.74	360	12
La ⁺⁺⁺	69.75	3.95	4.6	1.15	410	13-14

(The correction factor $\frac{r}{r_s}$ is read from Figure 6.1 for the value of r_s in the third column.)

THE VARIATION OF LIMITING IONIC CONDUCTIVITIES WITH TEMPERATURE

The data given in Appendix 6.1 are mainly confined to cases where accurate values of the limiting conductances of salts of the ions have been determined at 25° by means of measurements extending down to very low concentrations, *e.g.*, 10⁻³ to 10⁻⁴ N, making possible reliable extrapolations for Λ^0 . A few less reliable values, *e.g.*, that for Be⁺⁺, have been included for completeness. A much more extensive tabulation, including data for other temperatures, is given by WALDEN⁽⁸⁾. There is an extremely large body of conductivity data for 18°, which was the standard temperature for many physico-chemical studies in Britain and Europe until the 1920's, when the American practice of using 25° as the standard temperature became general. Other temperatures for which data are fairly plentiful are 0° and 100°. However, the use of the precise moving boundary method for determining transport numbers has been mainly confined to 25°, with the result that transport numbers for other temperatures are less certainly known.

The most accurate information we have on the variation of ionic mobilities with temperature comes from the work of GORDON and his collaborators^(1, 9), who have measured the conductivities and transport numbers of potassium chloride, sodium chloride and calcium chloride and the conductivity of potassium bromide at 15°,

VARIATION OF IONIC CONDUCTIVITIES WITH TEMPERATURE

25°, 35° and 45°. Their measurements were carried to concentrations low enough to permit reliable extrapolation to zero concentration.

Their transport numbers for the chloride ion in potassium chloride solution at infinite dilution, $t_{\text{Cl}^-}^0$, are plotted against the temperature in Figure 6.2. (The results do not confirm the claim of some earlier workers that transport numbers tend to approach the value 0.5 as the temperature is raised.) From this curve a value

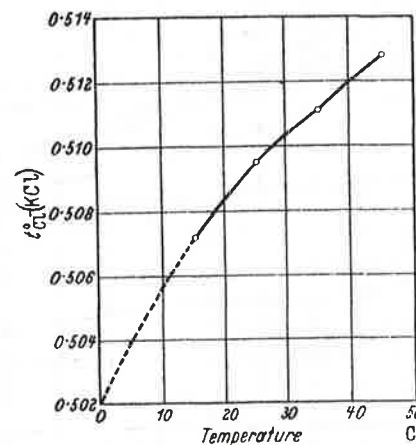


Figure 6.2. Limiting transport number of the chloride ion in aqueous potassium chloride as a function of temperature

of $t_{\text{Cl}^-}^0 = 0.5079$ may be interpolated at 18°. The four points on this graph suggest that an extrapolation to 0° would give $t_{\text{Cl}^-}^0 \approx 0.504$. Walden, before Gordon's data were available, estimated a value of 0.507. OWEN⁽¹⁰⁾ has fitted the conductivity data for a number of electrolytes to a cubic equation in the temperature; if his equation is used for extrapolation, it appears that at low temperatures the transport number decreases more rapidly with decreasing temperature than the results between 15° and 45° suggest. On this basis values of 0.502 and 0.504 at 0° and 5° respectively would seem reasonable and will be used for further calculations. At high temperatures the position is much less satisfactory. To extrapolate Gordon's data for more than 10° or so beyond 45° would be risky; but it seems likely that even at 100° $t_{\text{Cl}^-}^0$ should lie between 0.51 and 0.53. Owen's equations lead to 0.522; Walden in his compilation in Landolt-Börnstein's 'Tabellen' adopts the value 0.509; here we shall assume that $t_{\text{Cl}^-}^0(100^\circ) = 0.52$;

the difference is unimportant for the purpose of examining trends in ionic mobility with temperature, but our figure seems more in

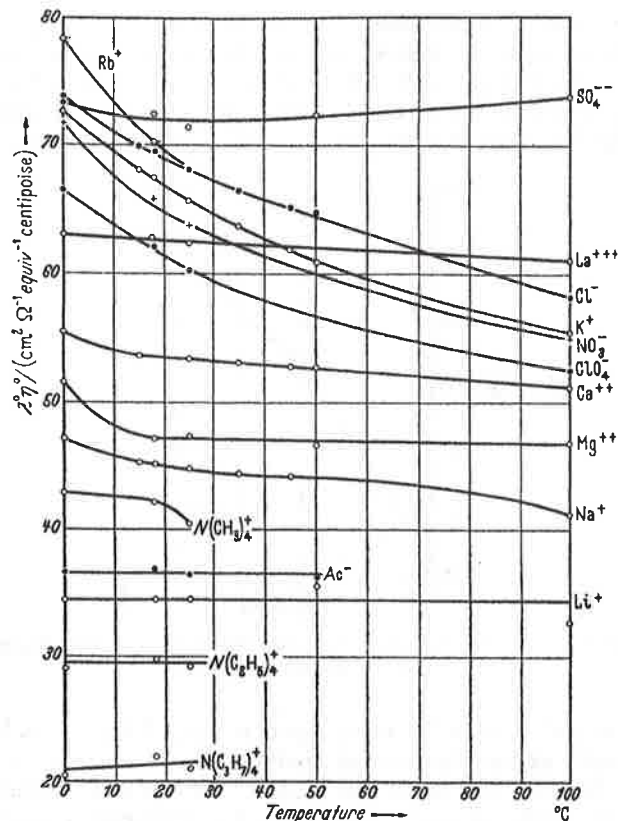


Figure 6.3. Variation of the product $\lambda^0\eta^0$ with temperature

keeping with the slow increase observed by Gordon between 15° and 45°.

Having settled upon these values for the transport number of chloride ion in potassium chloride, we can now use them in conjunction with the limiting conductivities of potassium chloride to calculate values for the limiting conductivity of the chloride ion at various temperatures.

With these values for the chloride ion as a basis, the limiting conductivities of many other single ions may be computed from the Λ^0 values for various salts. A representative selection is given in

Appendix 6.2 which has been compiled from the sources given in the footnotes.

The temperature variation of ionic conductivity is large, involving a five- or six-fold change over the range 0° to 100°. There is no doubt that the increasing mobility is closely related to the increasing fluidity of water; this can be shown by plotting the product $\lambda^0\eta^0$ (η^0 = the viscosity of water) against the temperature as in Figure 6.3. It is noteworthy that the ions for which $\lambda^0\eta^0$ is most nearly constant are those of large size, whether this large size is due to their being polyatomic (e.g., acetate and substituted ammonium ions) or to extensive hydration (e.g., Li^+ , Ca^{++} , La^{+++}). This observation does a good deal to justify the arguments by which we estimated the sizes of highly hydrated ions in terms of those of the substituted ammonium ions (see pp. 124-126).

The monatomic ions, K^+ , Rb^+ , Cl^- , Br^- , I^- , and ClO_4^- , NO_3^- are of similar mobility, and show a similar variation of the product $\lambda^0\eta^0$ with temperature; this behaviour shows up clearly in Figure 6.3 in contrast to the approximate constancy of $\lambda^0\eta^0$ for the larger ions. It should be noted, however, that even with these ions the variation of $\lambda^0\eta^0$ with temperature is only of the order of 30 per cent over the range 0° to 100°. This suggests that ordinary viscous forces account for most of the resistance to the motion of these ions in water, though there is evidently some other effect operative as well, which is important enough to render useless any attempt to estimate the sizes of these ions on the basis of Stokes' law.

IONIC MOBILITIES IN NON-AQUEOUS SOLVENTS

The measurement of the conductivities of non-aqueous solutions is a straightforward matter, the main requirements being careful attention to the purity of materials and the exclusion of atmospheric moisture. The work of KRAUS and his school⁽¹¹⁾ may be quoted as examples of the most precise techniques. It is, however, much more difficult to obtain from the experimental results accurate values for limiting ionic conductivities. First, the low dielectric constants of most non-aqueous solvents result in a much more pronounced decrease of the equivalent conductivity with concentration than is the case for aqueous solutions; and the theory required in extrapolating the conductivity to zero concentration is complicated by the effects of ion-pair formation. These difficulties can, however, be overcome, partly by carrying the measurements to very low concentrations, and partly by the introduction of a finite dissociation-constant into the conductivity formulae. The latter method

has been highly developed by Fuoss, and is discussed in Chapter 14.

The second and more serious difficulty is that at present there are practically no accurate transport number data available for non-aqueous electrolytes. The measurement of transport numbers in aqueous solutions has been developed to a high pitch of precision, the moving-boundary method of Longworth being the standard method for dilute solutions. GORDON and his co-workers⁽¹²⁾ have successfully applied this method to sodium and potassium chlorides in pure methanol and in methanol-water mixtures, (see Chapter 7) and HARNED and DREBY⁽¹³⁾ have derived the transport numbers of hydrochloric acid in dioxane-water mixtures from electromotive force studies. Measurements in mixed solvents, interesting though they are, present new theoretical problems connected with the preferential solvation of ions by one component of the solvent, and are therefore of less direct interest than would be the corresponding data in single non-aqueous solvents.

Many attempts have been made to estimate separate ionic conductivities from those of salts on a hydrodynamic basis. WALDEN⁽¹⁴⁾ found that the limiting equivalent conductivity of tetra-ethylammonium picrate in a wide variety of solvents including water conformed closely with the formula:

$$\Lambda^0\eta^0 = \text{constant}$$

which is derivable from Stokes' law and is known as Walden's rule. Walden's data showing the constancy of the product $\Lambda^0\eta^0$ for this particular salt at various temperatures and in various solvents are quite striking. A similar constancy was found for tetramethylammonium picrate, but the higher homologues, e.g., tetra-iso-amyl ammonium picrate, showed considerably larger variations of the $\Lambda^0\eta^0$ product. Walden considered that the constancy found for tetraethyl ammonium picrate justified the assumption that the picrate ion separately would have constant values of the product $\Lambda^0\eta^0$; on this basis, he deduced λ^0 values for the picrate ion in non-aqueous solvents from the known value of λ^0 in water, the only solvent for which accurate transport number measurements are available: for water, the value is found to be $\lambda^0\eta^0 = 0.270 \text{ cm}^2 \Omega^{-1} \text{ equiv}^{-1} \text{ poise}$. Limiting conductivities of other cations can then be found by subtraction of the appropriate value for the picrate ion from observed value of Λ^0 for various picrates in other solvents.

Walden's assumption of a constant value of $\lambda^0\eta^0$ for the picrate ion has been challenged by KRAUS⁽⁶⁾, who prefers to estimate transport numbers on the basis that the large tetra-*n*-butylammonium and triphenylborofluoride ions should have equal mobilities in

all solvents: this permits the estimation of separate ionic equivalent conductivities which are believed to be reliable within 5 per cent. The ionic conductivity values so obtained for the picrate and tetra-ethylammonium ions do not exhibit the constancy of the product $\lambda^0\eta^0$ assumed by Walden, that for the picrate ion varying from $\lambda^0\eta^0 \approx 0.24$ in ethylene dichloride to 0.30 in pyridine. Furthermore, Kraus cites data which show that the product $\Lambda^0\eta^0$ for tetra-ethylammonium picrate itself is by no means as constant as Walden claimed: this argument is independent of any arbitrary choice of transport numbers. In pyridine, in particular, the conductivity is abnormally high. While Walden's rule gives a useful guide to the conductivity to be expected, it cannot be considered quantitatively reliable. In a limited way, it is of value in interpreting the variation of ionic conductivities with temperature in one solvent: this aspect has been considered for aqueous ions on p. 124. The obtaining of accurate experimental transport number data for non-aqueous solutions is clearly the key to further progress in understanding the interactions of ions with these solvents.

The conductivities of electrolytes in methanol and hydrogen cyanide and some amides as solvents are discussed in the next chapter and concentrated sulphuric acid as a solvent for electrolytes is considered in Chapter 13. In solvents of lower dielectric constant, electrolytes readily form ion-pairs; this is discussed in Chapter 14, but mention may now be made of Appendix 14.2, which gives the limiting equivalent conductivities and dissociation constants of a number of electrolytes in non-aqueous solvents.

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THE VARIATION OF CONDUCTIVITIES AND TRANSPORT NUMBERS WITH CONCENTRATION

IN THE last chapter we have considered the equivalent and ionic conductivities at infinite dilution, *i.e.*, in a state where the ions are far enough apart to be without influence on one another. We now take up the question of the variation of conductivity with concentration, a problem which calls on all the resources of ionic interaction theory. There are two main effects of the interaction between the electric charges of the ions: these are the *electrophoretic effect* and the *relaxation effect*.

THE ELECTROPHORETIC EFFECT

The electrophoretic effect arises in the following way. When an ion moves through a viscous medium it tends to drag along with it the solution in its vicinity. Neighbouring ions therefore have to move not in a stationary medium but with or against the stream according as they are moving in the same direction as the first ion or oppositely. The effect will clearly be concentration-dependent, falling to zero at infinite dilution, and its computation will require the use of the distribution function, since it involves the distances between ions. For the equilibrium case where no external forces such as electric fields or concentration-gradients are acting on the solution, we have been obliged to adopt the distribution functions (4.9) for unsymmetrical electrolytes, and (4.14) for symmetrical electrolytes; these conform to the Boltzmann distribution law as nearly as is permitted by the principle of the linear superposition of fields, and are mathematically consistent with the expression (4.13) for the potential ψ_j . If the ions are moving under the influence of external forces, these distributions will in general be disturbed. In the case of the diffusion of a single electrolyte, however, all the ions must move with the same velocity, and the symmetry of the distribution is not affected. In this case, therefore, the electrophoretic effect may legitimately be computed from these distribution functions. In electrical conduction the symmetry will be disturbed; this gives rise to the relaxation effect which will be

discussed later, but we shall neglect the effect of the dissymmetry when calculating the electrophoretic effect in conduction. Also, for the sake of generality, we shall use the Boltzmann distribution law (4.5) rather than forms (4.9) or (4.14), in order to facilitate the investigation of questions of convergence; the results corresponding to the distributions (4.9) and (4.14) can then be obtained as special cases of the general formula. However, we retain the simple expression (4.13) for the potential ψ . The treatment of electrophoresis given here is essentially that of ONSAGER and FUOSS⁽¹⁾, but employs the general term of the Boltzmann distribution law. We shall also confine ourselves to solutions containing only a single electrolyte, subscripts 1 and 2 denoting cations and anions respectively, and a subscript A the solvent.

In the cases considered here, bulk motion of the solution as a whole is irrelevant; it follows that the forces k_1 and k_2 acting on the ions must be balanced by other forces k_A acting on the solvent molecules; and denoting the respective bulk concentrations by n_1 , n_2 and n_A , we have:

$$n_A k_A = -n_1 k_1 - n_2 k_2 \quad \dots (7.1)$$

At a distance r from a chosen cation the local concentrations of ions are given by the Boltzmann expression (4.5). A spherical shell of radius r and thickness dr is subject to a resultant force given by:

$$(n'_1 k_1 + n'_2 k_2 + n_A k_A) 4\pi r^2 dr$$

Provided that we neglect any variation in n_A at this point from its bulk value (a course which is safe for dilute solutions), we can eliminate $n_A k_A$ by means of (7.1) obtaining for the resultant force on the shell:

$$[(n'_1 - n_1)k_1 + (n'_2 - n_2)k_2] 4\pi r^2 dr$$

This force is assumed to cause the shell and all points within it to move with a velocity obtained, according to Stokes' law, by dividing the force by $6\pi\eta r$. Each shell thus contributes an electrophoretic increment to the velocity of the central ion, and the whole increment, Δv_1 , is obtained by integrating over all the shells, beginning at $r = a$ (the distance within which no other ions can penetrate and within which the electrophoretic velocity remains constant). This gives:

$$\Delta v_1 = \frac{2}{3\eta} \int_{r=a}^{\infty} [(n'_1 - n_1)k_1 + (n'_2 - n_2)k_2] r dr \quad \dots (7.2)$$

If we take n'_1 and n'_2 to be given by the Boltzmann distribution (4.5), we have on expanding the exponentials:

$$\begin{aligned} n'_1 - n_1 &= n_1 \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left(\frac{z_1 e\psi}{kT} \right)^n \\ n'_2 - n_2 &= n_2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left(\frac{z_2 e\psi}{kT} \right)^n \end{aligned} \quad \dots (7.3)$$

It will prove convenient to express the concentrations n_1 and n_2 in these formulae in terms of the quantity (κa) . From equation (4.12) we have, using the electrical neutrality condition, $n_1 z_1 + n_2 z_2 = 0$,

$$\begin{aligned} n_1 &= \frac{(\kappa a)^2}{4\pi a^2} \left(\frac{e^2}{\epsilon kT} \right)^{-1} \frac{1}{z_1(z_1 - z_2)} \\ n_2 &= \frac{(\kappa a)^2}{4\pi a^2} \left(\frac{e^2}{\epsilon kT} \right)^{-1} \frac{1}{z_2(z_2 - z_1)} \end{aligned}$$

Using these values in (7.3) and taking the potential ψ to be given by equation (4.13), we obtain from equation (7.2):

$$\begin{aligned} \Delta v_1 &= \frac{1}{6\pi\eta} \frac{(\kappa a)^2}{a^2} \left(\frac{e^2}{\epsilon kT} \right)^{-1} \sum_{n=1}^{\infty} \left[z_1^n \frac{z_1^{n-1} k_1 - z_2^{n-1} k_2}{z_1 - z_2} \cdot \frac{(-1)^n}{n!} \right. \\ &\quad \left. \cdot \left(\frac{e^2}{\epsilon kT} \right)^n \left(\frac{e^{\kappa a}}{1 + \kappa a} \right)^n \int_a^{\infty} \frac{e^{-n\kappa r}}{r^{n-1}} dr \right] \quad \dots (7.4) \end{aligned}$$

The integral occurring in (7.4) can always be evaluated as:*

$$\int_a^{\infty} \frac{e^{-n\kappa r}}{r^{n-1}} dr = \frac{S_n(\kappa a)}{a^{n-2}} \quad \dots (7.5)$$

where $S_n(\kappa a)$ is a function of (κa) only. Equation (7.4) may therefore be more briefly expressed as:

$$\Delta v_1 = \sum_{n=1}^{\infty} \frac{(-1)^n}{6\pi\eta n!} \left(\frac{e^2}{\epsilon kT} \right)^{n-1} \frac{z_1^n}{a^n} \frac{z_1^{n-1} k_1 - z_2^{n-1} k_2}{z_1 - z_2} \phi_n(\kappa a) \quad \dots (7.6)$$

where the function $\phi_n(\kappa a)$ is a function of (κa) only, and is defined by:

$$\phi_n(\kappa a) = (\kappa a)^2 \left(\frac{e^{\kappa a}}{1 + \kappa a} \right)^n S_n(\kappa a)$$

* See appendix to this chapter, p. 170.

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The corresponding equation for the electrophoretic increment to the velocity of an anion, Δv_2 , is obtained by merely interchanging the subscripts 1 and 2 throughout equation (7.6). The further abbreviation:

$$\Delta v_1 = \sum_{n=1}^{\infty} A_n \cdot \frac{z_1^n (z_1^{n-1} k_1 - z_2^{n-1} k_2)}{a^n (z_1 - z_2)} \quad \dots (7.7)$$

is useful where A_n depends only on (κa) , temperature, and solvent properties and is given by:

$$A_n = \frac{(-1)^n}{n! 6\pi\eta} \left(\frac{e^2}{\epsilon k T} \right)^{n-1} \phi_n(\kappa a) \quad \dots (7.8)$$

Equation (7.7) contains the (as yet) unspecified forces k_1 and k_2 which act on the ions. In conductance, these forces are given by the product of the field intensity and the ionic charge; in diffusion, they are the combination of a virtual force produced by the gradient of chemical potential, and an electrical force due to the 'diffusion potential' which results from the electrical attraction of the faster moving for the slower moving ionic species. The application of equation (7.7) to these phenomena will be discussed later.

THE 'RELAXATION EFFECT' IN CONDUCTIVITY

In general, the motion of ions under the influence of external forces will disturb the symmetrical distribution of the ions, and one would therefore expect that this disturbance would tend to decrease the velocity of the ions. In the solution in equilibrium, the 'ionic atmosphere' (which is a convenient description of the whole assemblage of ions outside the central one chosen) is on a time-average distributed with spherical symmetry, and therefore exerts no resultant force on the central ion. The central ion may then be pictured as moving to an off-centre position and experiencing a restoring force, which, however, rapidly dies away as the 'atmosphere' is rearranged by the thermal motions of its constituent ions. The molecular picture thus involves the concept of the 'relaxation of the ionic atmosphere' and the average restoring force experienced by the ion is called the relaxation effect. The external force acting on the ion may, in the conductivity problem, be taken as a field of intensity X acting in the x -direction; the 'relaxation field' will clearly act in the same direction but in the opposite sense, and will be denoted by ΔX . The computation of ΔX involves a combination of the ideas of the interionic attraction theory with the equation of continuity of hydrodynamics, and is mathematically the most

THE 'RELAXATION EFFECT' IN CONDUCTIVITY

difficult part of electrolyte theory. Because of the unavoidable complexity of the treatment we shall not give the complete derivation here, but shall merely give a statement of the main results.

The first attack on the problem of the relaxation effect was made by DEBYE and HÜCKEL⁽²⁾; a more successful approach, however, was that of ONSAGER⁽³⁾, who obtained the following limiting law for the relaxation effect on the conductivity of an extremely dilute solution of a single electrolyte dissociating into ions 1 and 2:

$$\frac{\Delta X}{X} = \frac{z_1 z_2 e^2}{3 \epsilon k T} \cdot \frac{q \kappa}{1 + \sqrt{q}} \quad \dots (7.9)$$

Here the quantity q is defined by:

$$\begin{aligned} q &= \frac{|z_1 z_2|}{|z_1| + |z_2|} \cdot \frac{\lambda_1^0 + \lambda_2^0}{|z_2| \lambda_1^0 + |z_1| \lambda_2^0} \\ &= \frac{|z_1 z_2|}{(|z_1| + |z_2|)(|z_2| \lambda_1^0 + |z_1| \lambda_2^0)} \\ &= \frac{1}{2} \text{ for symmetrical electrolytes where } |z_1| = |z_2| \quad \dots (7.10) \end{aligned}$$

The total electric force acting on the ion is thus given by $X z_j e$ $\left(1 + \frac{\Delta X}{X}\right)$ and produces a velocity (relative to the solvent) of:

$$v_j' = X z_j e u_j^0 \left(1 + \frac{\Delta X}{X}\right) \quad \dots (7.11)$$

where u_j^0 is the absolute mobility of the ion. At infinite dilution, the velocity produced by the field X is:

$$v_j^0 = X z_j e u_j^0 \quad \dots (7.12)$$

Hence introducing (7.9) we have:

$$v_j' = v_j^0 \left(1 + \frac{z_1 z_2 e^2}{3 \epsilon k T} \cdot \frac{q \kappa}{1 + \sqrt{q}}\right)$$

as Onsager's expression for the velocity of the ion, corrected for the relaxation effect. Before calculating the further correction required to take the electrophoretic effect into account, we shall consider later developments in the theory of the relaxation effect. In Onsager's treatment, several approximations are made: (a) The potential ψ_j is taken as given by the expression:

$$\psi_j = \frac{z_j e}{\epsilon} \frac{e^{-\kappa r}}{r}$$

i.e., the factor $\frac{e^{\kappa a}}{1 + \kappa a}$ is omitted from equation (4.13). This means that the resulting expression is valid only at great dilutions where κa is small compared with unity. (b) Various other approximations involving the relation $\Delta X \ll X$ are made: these also will be admissible at great dilutions where the relaxation effect is small.

For some twenty-five years after the appearance of Onsager's theory no major progress was made with the relaxation-effect. From 1952 onwards however a number of extensions and refinements have appeared, in all of which the most important new feature is the introduction of the ion size parameter a , making possible a considerable increase in the range of validity of the theory.

The 1952 paper of FALKENHAGEN, LEIST and KELBG⁽⁴⁾ employed the Eigen-Wicke distribution function mentioned in Chapter 4, in place of the usual Boltzmann one; this makes only a minor difference in the meaning of the quantity κ and need not be considered further here. They obtained for the relaxation-effect, allowing for finite ion size, the expression:

$$\frac{\Delta X}{X} = \frac{z_1 z_2 e^2}{3\epsilon k T} \cdot \frac{q}{1 - q} \cdot \frac{\kappa}{(1 + \kappa a)\kappa a} [e^{\kappa a(1 - \sqrt{q})} - 1] \quad \dots (7.13)$$

Expanding the exponential in (7.13) as far as the first power of (κa) gives:

$$\frac{\Delta X}{X} = \frac{z_1 z_2 e^2}{3\epsilon k T} \cdot \frac{q}{1 + \sqrt{q}} \cdot \frac{\kappa}{1 + \kappa a} \quad \dots (7.14)$$

which differs from Onsager's result (7.9) only by the factor $(1 + \kappa a)$ in the denominator. Thus they found the effect of finite ion size on the relaxation-effect to be of the same form as its effects on the free energy and on the electrophoretic effect (cf. equation 9.5 and 7.27).

Almost simultaneously, PITTS⁽⁵⁾ investigated the conductance of symmetrical electrolytes; his equation is compared with others in reference (6). His result, on separating out the part dealing with the relaxation-effect, may be written:

$$-\frac{\Delta X}{X} = \frac{z^2 e^2}{3\epsilon k T} \cdot \frac{q}{1 + \sqrt{q}} \cdot \frac{\kappa}{(1 + \kappa a)(1 + \kappa a \sqrt{q})} + \left(\frac{z^2 e^2 \kappa}{\epsilon k T} \right)^2 \frac{S_1}{3} \quad \dots (7.15)$$

The second term on the right arises from the 'higher terms', S_1 being given as a function of (κa) in a table in the original paper. Ignoring this term, which will not appear if we adopt the 'self-consistent'

treatment of the potential problem as discussed in Chapter 4, we see that (7.15) differs from (7.14) by the presence of a further factor $(1 + \kappa a \sqrt{q})$ in the denominator.

In a subsequent paper⁽⁷⁾, Falkenhagen's school introduced another boundary condition, that the normal component of the relative motion of two ions must vanish at the surface of ions in contact since they are treated as hard spheres. On this basis they evaluated the relaxation-effect as:

$$\frac{\Delta X}{X} = -\frac{e^2}{3\epsilon k T} \cdot \frac{q}{1 + \sqrt{q}} \cdot \frac{\kappa}{(1 + \kappa a)[1 + \kappa a \sqrt{q} + \kappa^2 a^2/6]} \quad \dots (7.16)$$

for 1:1 electrolytes, with $q = 0.5$. This equation is identical with the first-order term of Pitts's result (7.15), except for the further term $(\kappa a)^2/6$ in the denominator.

In 1953 MIRTSKHULAVA⁽⁸⁾ also published a treatment of the problem along the same lines as Pitts. Her result for the relaxation-effect is given as a complicated power series (equation 38 of ref. 8), involving also a term containing the exponential integral function. The latter is of especial interest since at very low concentrations it gives rise to a term in $(c \log c)$, as had been anticipated by ONSAGER and FUOSS⁽¹⁾.

FUOSS and ONSAGER⁽⁹⁾ have recently given the most comprehensive treatment, including numerical tables of certain transcendental functions related to the exponential integral functions. Their final result, giving the contributions of both the relaxation effect and the electrophoretic effect, is expressed as Onsager's original limiting-law result together with a very complicated function of κ and a , for which the original papers must be consulted. They demonstrate that the transcendental functions involved lead to a term of order $(c \log c)$, although the approximations giving this form are valid only at extremely low concentrations. This contribution from the transcendental terms is quite small, but its relative importance increases at low concentrations, and Fuoss and Onsager show that neglect of it can lead to small errors in the extrapolation of conductance data. They emphasize that the approximation of their transcendental terms to the form $(c \log c)$ at low concentrations does not justify the use of such terms with arbitrary coefficients for the representation of conductance data at higher concentrations, a practice which has been followed by many workers in the past. An arduous but worthwhile task for an enthusiastic algebraist would be to determine to what extent the formulae of Fuoss and Onsager agree with

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those of Mirtskhulava, which as far as we know have not yet been tested against experimental results.

We do not consider it practicable to present the Fuoss-Onsager treatment in detail; even the original papers give only a condensed account of the development of the formulae. We shall therefore use for illustrating the general form of the theory, Falkenhagen's expression (7.16) for the relaxation effect.

THE EFFECT OF ELECTROPHORESIS ON THE CONDUCTIVITY

The general equation (7.7) for the electrophoretic increment to the ionic velocity may now be specialized for the case of conductivity by replacing the forces k_1 and k_2 , which act on the ions, by the sum of the forces produced by the external field X and the relaxation-field ΔX giving:

$$k_1 = (X + \Delta X)z_1e, \quad k_2 = (X + \Delta X)z_2e \quad \dots (7.17)$$

Equation (7.7) then becomes:

$$\left. \begin{aligned} \Delta v_1 &= (X + \Delta X)e \sum_n A_n \frac{z_1^{2n} - z_1^n z_2^n}{a^n(z_1 - z_2)} \\ \Delta v_2 &= (X + \Delta X)e \sum_n A_n \frac{z_1^n z_2^n - z_2^{2n}}{a^n(z_1 - z_2)} \end{aligned} \right\} \quad \dots (7.18)$$

Hence the final velocity of the ions, corrected for both electrophoretic and relaxation effects, is given by combining equations (7.11) and (7.18):

$$\begin{aligned} v_1 &= v_1' + \Delta v_1 \\ &= (X + \Delta X)z_1eu_1^0 + (X + \Delta X)e \sum_n A_n \frac{z_1^{2n} - z_1^n z_2^n}{a^n(z_1 - z_2)} \quad \dots (7.19) \end{aligned}$$

But the absolute mobility u_1^0 is also given by equation (7.12) in terms of the velocity v_1^0 produced by the field X at infinite dilution:

$$v_1^0 = Xz_1eu_1^0 \quad \dots (7.20)$$

Dividing equation (7.19) by (7.20) gives:

$$\frac{v_1}{v_1^0} = \left(1 + \frac{\Delta X}{X}\right) \left[1 + \frac{1}{z_1u_1^0} \sum_n A_n \frac{z_1^{2n} - z_1^n z_2^n}{a^n(z_1 - z_2)}\right] \quad \dots (7.21)$$

Since the velocities v_1 and v_1^0 are those attained under the same

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external field X in the actual solution and in the infinitely dilute solution respectively, the ratio v_1/v_1^0 may be replaced by the ratio of the equivalent ionic conductivities λ_1/λ_1^0 ; and the factor $\frac{1}{z_1u_1^0}$ preceding the summation on the right may be put in terms of λ_1^0 , by the relation (cf. 2.46):

$$u_1^0 = N\lambda_1^0/(F^2|z_1|)$$

so that (7.21) becomes:

$$\lambda_1 = \left(\lambda_1^0 + \frac{F^2}{N} \sum_n A_n \frac{z_1^{2n} - z_1^n z_2^n}{a^n(|z_1| + |z_2|)}\right) \left(1 + \frac{\Delta X}{X}\right) \quad \dots (7.22)$$

the relaxation term $\frac{\Delta X}{X}$ being given by equations (7.9), (7.14) or (7.16) according to the degree of approximation desired. The corresponding expression for the anion is:

$$\lambda_2 = \left(\lambda_2^0 + \frac{F^2}{N} \sum_n A_n \frac{z_2^{2n} - z_1^n z_2^n}{a^n(|z_1| + |z_2|)}\right) \left(1 + \frac{\Delta X}{X}\right) \quad \dots (7.23)$$

The equivalent conductivity of the electrolyte $\Lambda = \lambda_1 + \lambda_2$, is therefore:

$$\Lambda = \left(\Lambda^0 + \frac{F^2}{N} \sum_n A_n \frac{(z_1^n - z_2^n)^2}{a^n(|z_1| + |z_2|)}\right) \left(1 + \frac{\Delta X}{X}\right) \quad \dots (7.24)$$

Though we have retained the general expression⁽¹⁰⁾ for the electrophoretic terms in developing these expressions, it will be recalled that the Boltzmann distribution on which this expression is based is not mathematically consistent with the Poisson equation, and that for consistency the series can be taken only as far as the first term for unsymmetrical valency types, and the second for symmetrical types. Furthermore, it is obvious from equations (7.22), (7.23) and (7.24) that in the case of symmetrical electrolytes, ($z_1 = -z_2$), the second-order electrophoretic term ($n=2$) vanishes identically. Hence, in all cases the first-order term alone need really be considered though examination of the convergence of higher-order terms may throw useful light on the validity of the approximation made to the Boltzmann distribution.

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The first-order electrophoretic term in the conductivity equation

Taking only the term for $n = 1$ and using the definitions of A_n , ϕ_n and S_n , given on pp. 135–136, we have:

$$A_1 = -\frac{1}{6\pi\eta} \cdot \frac{\kappa a}{1 + \kappa a}$$

Equations (7.22), (7.23) and (7.24) therefore become:

$$\lambda_1 = \left(\lambda_1^0 - \frac{F^2}{6\pi\eta N} |z_1| \frac{\kappa}{1 + \kappa a} \right) \left(1 + \frac{\Delta X}{X} \right) \dots (7.25)$$

$$\lambda_2 = \left(\lambda_2^0 - \frac{F^2}{6\pi\eta N} |z_2| \frac{\kappa}{1 + \kappa a} \right) \left(1 + \frac{\Delta X}{X} \right) \dots (7.26)$$

$$\Lambda = \left(\Lambda^0 - \frac{F^2}{6\pi\eta N} (|z_1| + |z_2|) \frac{\kappa}{1 + \kappa a} \right) \left(1 + \frac{\Delta X}{X} \right) \dots (7.27)$$

THE ONSAGER LIMITING LAW FOR THE CONDUCTIVITY

In Onsager's treatment, the further approximation of writing $(1 + \kappa a) \approx 1$ in the denominator of the first-order electrophoretic correction is made, and the relaxation term $\frac{\Delta X}{X}$ is expressed by the limiting equation (7.9). Further, in evaluating the electrophoretic effect the forces k_1 and k_2 are taken as Xz_1e and Xz_2e rather than as $(X + \Delta X)z_1e$ and $(X + \Delta X)z_2e$; this is equivalent to neglecting the cross-product of the electrophoretic and relaxation terms in (7.25)–(7.27). All these approximations are of course quite justified for the purpose of finding the limiting law, but it is clear that the resulting expression will apply only at extreme dilutions, since (κa) is far from negligible compared to unity at ordinary concentrations, and diminishes only as $\sqrt{c}$. The Onsager limiting law is thus:

$$\Lambda = \Lambda^0 - \frac{|z_1 z_2| e^2}{3\epsilon k T} \cdot \frac{\Lambda^0 q \kappa}{1 + \sqrt{q}} - \frac{F^2}{6\pi\eta N} (|z_1| + |z_2|) \kappa \dots (7.28)$$

Since κ is given by equation (4.12) it may be written as:

$$\kappa = \left(\frac{8\pi N e^2}{1000 \epsilon k T} \right)^{\frac{1}{2}} \sqrt{I}$$

CONDUCTIVITY EQUATIONS FOR HIGHER CONCENTRATIONS

where I is the 'ionic strength' defined by:

$$I = \frac{c}{2} (\nu_1 z_1^2 + \nu_2 z_2^2)$$

with the usual convention that c implies the mole/l scale. Equation (7.28) then becomes, on inserting the values of the physical constants:

$$\Lambda = \Lambda^0 - \left[\frac{2.801 \times 10^6 |z_1 z_2| q \Lambda^0}{(\epsilon T)^{3/2} (1 + \sqrt{q})} + \frac{41.25 (|z_1| + |z_2|)}{\eta (\epsilon T)^{\frac{1}{2}}} \right] \sqrt{I} \dots (7.29)$$

in which η must be expressed in poise and T in $^{\circ}K$.

This limiting law is of the form:

$$\Lambda = \Lambda^0 - A \sqrt{c} \dots (7.30)$$

which was found by Kohlrausch to describe the variation of equivalent conductivity with concentration in dilute solutions. For aqueous solutions at 25° it reduces on putting $\epsilon = 78.30$, $T = 298.16^{\circ}K$ and $\eta = 0.008903$ poise, with Λ expressed as $(cm^2 \Omega^{-1} equiv^{-1})$, to:

$$\Lambda = \Lambda^0 - \left[0.7852 |z_1 z_2| \frac{q \Lambda^0}{1 + \sqrt{q}} + 30.32 (|z_1| + |z_2|) \right] \sqrt{I} \dots (7.31)$$

CONDUCTIVITY EQUATIONS FOR HIGHER CONCENTRATIONS

For many years equation (7.29) was employed with added terms in c , $c^{3/2}$, $c \log c$, etc., to represent data at concentrations above $0.001N$, where Onsager's limiting form is no longer adequate. For purposes of extrapolation, an equation proposed by SHEDLOVSKY⁽¹¹⁾ has been widely used: equation (7.29) may be written:

$$\Lambda = \Lambda^0 - (B_1 \Lambda^0 + B_2) \sqrt{c}$$

where B_1 and B_2 are parameters given by the theory. Rearranging this to:

$$\Lambda^0 = (\Lambda + B_2 \sqrt{c}) / (1 - B_1 \sqrt{c}), \dots (7.32)$$

Shedlovsky observed that for strong aqueous 1:1 electrolytes the quantity on the right of (7.32) is not constant, as it would be if equation (7.29) were obeyed exactly, but varies almost linearly

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with c up to concentrations of about 0.1N. He therefore defined an extrapolation function, Λ^0 , by:

$$\Lambda^0 = (\Lambda + B_2\sqrt{c})/(1 - B_1\sqrt{c}) \quad \dots(7.33)$$

which when plotted against c would yield on extrapolation to $c = 0$, the true limiting conductivity Λ^0 . This implies that data up to 0.1N can be fitted by the expression:

$$\Lambda = \Lambda^0 - (B_1\Lambda^0 + B_2)\sqrt{c} + b c(1 - B_1\sqrt{c}) \quad \dots(7.34)$$

where the coefficient b is chosen to fit the data. Useful as this device is, its weakness lies in its empirical nature, no simple meaning being attached to the b coefficient. FUOSS and ONSAGER⁽⁹⁾ have shown that the approximate constancy of b can be accounted for as a fortuitous consequence of the numerical values of certain terms in their complete theory.

Still denoting the coefficients of the relaxation and electrophoretic terms in equation (7.29) by B_1 and B_2 respectively, we can combine Falkenhagen's equation (7.16) with equation (7.27) to obtain:

$$\Lambda = \left(\Lambda^0 - \frac{B_2\sqrt{c}}{1 + \kappa a} \right) \left(1 - \frac{B_1\sqrt{c}}{(1 + \kappa a)(1 + \kappa a\sqrt{q} + \kappa^2 a^2/6)} \right) \quad \dots(7.35)$$

If we expand this product in powers of $\sqrt{c}$, putting $\kappa a = Ba\sqrt{c}$, we obtain:

$$\Lambda = \Lambda^0 - (B_1\Lambda^0 + B_2)\sqrt{c} + c(aBB_2 + B_1B_2 + 1.707\Lambda^0 a BB_1) - 2.707 a BB_1 B_2 c^{3/2} + \dots$$

Since for most aqueous 1:1 electrolytes $Ba \approx 1$ (mole^{-1/2} litre^{1/2}), whilst $B_1 \approx 0.2$, $B_2 \approx 60$ and $\Lambda \approx 100$, the coefficients of the terms in c and $B_1 c^{3/2}$ are of the same magnitude; to this extent the expression provides some justification for the form of Shedlovsky's function.

Another useful approximation is that proposed by the authors⁽¹²⁾ shortly after the appearance of Falkenhagen's earlier equation (7.13). Combining (7.14) with (7.27) and neglecting the cross-product of the relaxation and electrophoretic terms, gives:

$$\Lambda = \Lambda^0 - \frac{B_1\Lambda^0 + B_2}{1 + \kappa a} \sqrt{c} \quad \dots(7.36)$$

i.e., we have merely to divide the square-root term of Onsager's original limiting law by $(1 + \kappa a)$ in order to allow for the finite

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ionic size. Equation (7.36) gives a very fair account of the conductances of aqueous 1:1 electrolytes up to 0.05 or 0.1N. It may be rearranged as:

$$\Lambda^0 = \Lambda + \frac{B_1\Lambda + B_2}{1 + (Ba - B_1)\sqrt{c}} \sqrt{c} \quad \dots(7.37)$$

Table 7.1

Values of the function $\phi_n(\kappa a) = (\kappa a)^2 \left(\frac{e^{\kappa a}}{1 + \kappa a} \right)^n S_n(\kappa a)$

at round (κa) where $S_n(\kappa a) = a^{n-2} \int_a^\infty \frac{e^{-\kappa r}}{r^{n-1}} dr$

κa	$100\phi_1(\kappa a)$	$100\phi_2(\kappa a)$	$100\phi_3(\kappa a)$	$100\phi_4(\kappa a)$	$100\phi_5(\kappa a)$
0.0	0	0	0	0	0
0.05	4.762	0.4566	0.1609	0.0884	0.0584
0.1	9.091	1.235	0.4761	0.2624	0.1693
0.2	16.67	2.911	1.166	0.6192	0.3764
0.3	23.08	4.405	1.734	0.876	0.499
0.5	33.33	6.628	2.425	1.100	0.552
0.7	41.18	7.987	2.686	1.096	0.491
1.0	50.00	9.032	2.678	0.938	0.37
1.2	54.55	9.327			
1.4	58.33	9.434			
1.5	60.00	9.429			
1.6	61.54	9.411			
1.8	64.29	9.316			
2.0	66.67	9.170			
2.5	71.43	8.692			
3.0	75.00	8.172			
3.5	77.78	7.662			
4.0	80.00	7.187			
4.5	81.82	6.753			
5.0	83.33	6.359			
5.5	84.62	6.002			
6.0	85.71	5.681			

From STOKES, R. H., *J. Amer. chem. Soc.*, 75 (1953) 4563.

where $\kappa = B\sqrt{c}$, which is a useful form for determining Λ^0 . Its advantage over Shedlovsky's function is that the parameter a has a simple physical meaning and can be expected to lie in the range 3 - 5.5Å; for fully dissociated 1:1 electrolytes it has been found to be nearly independent of temperature for any given electrolyte (see Table 7.3).

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To each of these equations (7.29, 7.35, 7.36) for Λ there of course corresponds a pair of equations for λ_1 and λ_2 separately; these are obtained from equations (7.25) and (7.26) by the same method, and differ only in that λ_1 or λ_2 replaces Λ , λ_1^0 or λ_2^0 replaces Λ^0 , and in the electrophoretic term $|z_1|$ or $|z_2|$ replaces the sum $(|z_1| + |z_2|)$. The relaxation term for the separate ions in the equations corresponding to (7.33) and (7.34) is exactly the same as in these equations for Λ .

CONVERGENCE OF THE ELECTROPHORETIC TERMS

We now return to equation (7.24) where the electrophoretic contribution appears as the series:

$$\frac{F^2}{N} \sum_n A_n \frac{(z_1^n - z_2^n)^2}{a^n(|z_1| + |z_2|)}$$

In this series the quantities A_n are given by the equation:

$$A_n = \frac{(-1)^n}{n!6\pi\eta} \left(\frac{e^2}{\epsilon k T} \right)^{n-1} \phi_n(\kappa a)$$

The dimensions of A_n are those of (viscosity⁻¹ lengthⁿ⁻¹). The dimensionless function $\phi_n(\kappa a)$ for values of n up to 5 is given in Table 7.1. In deriving the conductivity equations, we have used only the first-order electrophoretic term, obtained by putting $n = 1$

Table 7.2
Values of the valency factor $\frac{(z_1^n - z_2^n)^2}{a^n(|z_1| + |z_2|)}$

Valency type	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
1 : 1	$2/a$	0	$2/a^3$	0	$2/a^5$
2 : 2	$4/a$	0	$64/a^3$	0	$1024/a^5$
1 : 2 and 2 : 1	$3/a$	$3/a^2$	$27/a^3$	$75/a^4$	$363/a^5$

in the above formula, a course adopted in the interest of self-consistency. The question as to whether the series does converge rapidly enough to make this expedient successful has been investigated by STOKES⁽¹⁰⁾ who showed that for aqueous solutions at 25° convergence depends on the factor:

$$\frac{(z_1^n - z_2^n)^2}{a^n(|z_1| + |z_2|)}$$

rather than the quantity A_n . Table 7.2 shows how this factor behaves for various valency types.

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Bearing in mind that the ion-size parameter a is in the vicinity of 4Å for most simple ions, it is apparent from Table 7.2 that the electrophoretic terms given by the series in (7.24) will converge quite satisfactorily for 1 : 1 electrolytes in water, so that the formulae proposed for the conductivity, in which only the term $n = 1$ is accepted, should be adequate. For 2 : 2 electrolytes, in spite of the vanishing of the even-order terms, the convergence is unsatisfactory since the third- and fifth-order terms will be of comparable magnitude to the first. For unsymmetrical valency types, none of the terms vanish and convergence will be slow, though somewhat assisted by the alternation of signs. In non-aqueous solvents the dielectric constant is usually lower than that of water (hydrocyanic acid is one exception); therefore the factor $\frac{e^2}{\epsilon k T}$ which appears to the $(n - 1)th$ power in the expression for A_n , will be larger and will militate further against satisfactory convergence. We cannot, therefore, really expect the present treatment to be quantitatively successful except for 1 : 1 electrolytes in water, unless there are good reasons for ascribing a large effective size a to the ions; this foreboding is, in fact, fulfilled. Merely to include the higher-order electrophoretic terms is not a satisfactory solution, since the formulae from which these terms have been computed are self-consistent only as far as the first order for unsymmetrical electrolytes and as far as the second order for symmetrical ones. The formulae obtained above for the higher-order terms are therefore helpful only when these are negligible; in other cases they serve to demonstrate the inadequacy of the treatment rather than to provide an adequate one.

EXPERIMENTAL TESTS OF THE THEORY OF CONDUCTIVITY

The Onsager limiting law (7.29) has been exhaustively tested by extremely precise experimental studies, and its validity, for the conditions assumed in its derivation, has been conclusively demonstrated. These conditions may be summed up by the requirements that the dimensionless parameter (κa) should be very small compared to unity, and that the electrolyte should be fully dissociated into ions. For aqueous solutions at ordinary temperatures, κ is approximately $0.3 \times 10^8 \sqrt{I}$ (see Appendix 7.1); and the mean ionic diameter a is $3-5 \times 10^{-8}$ cm; hence κa is of the same order of magnitude as $\sqrt{I}$. At a concentration where the ionic strength I is 0.001, κa is about 0.03, and the approximation of neglecting it in the factor $(1 + \kappa a)$ therefore involves an error of about 3 per cent in the value of $(\Lambda^0 - \Lambda)$. At this ionic strength, $(\Lambda^0 - \Lambda)$ is

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about three equivalent conductivity units for 1 : 1 electrolytes, and rather more for higher valency types, so that the approximation is of the order of 0.1 units in Λ . This is several times the experimental error of the best work and shows that a concentration of one-thousandth normal must be regarded as a theoretical upper bound to the range of validity of the limiting law, even for aqueous 1 : 1 electrolytes. In other solvents and for higher valency-types the

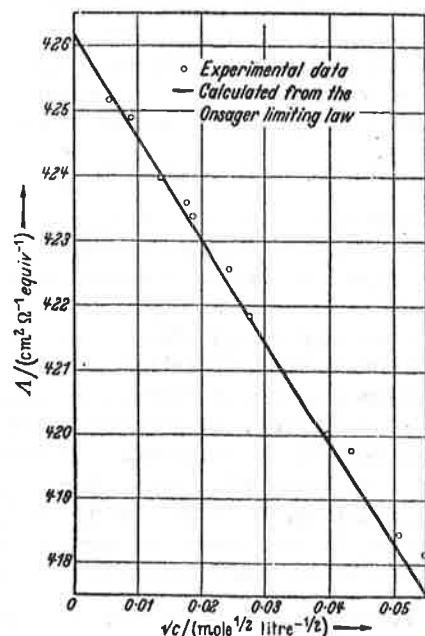


Figure 7.1. Equivalent conductivity of hydrochloric acid in very dilute aqueous solution at 25°

bound is even lower. However, conductivity measurements can in favourable cases be carried out accurately at concentrations as low as 0.00003 N: although the measured values at such concentrations are often close to the limiting value for infinite dilution, the experimental precision is such that the difference still makes a significant test of the limiting law. The very careful measurements of SHEDLOVSKY *et al.*⁽¹³⁾ have shown that in the concentration-range from 0.00003 to 0.001 N the Onsager formula (7.29) is obeyed within experimental error by aqueous sodium chloride, potassium chloride, hydrochloric acid, silver nitrate, calcium chloride and lanthanum chloride.

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Silver nitrate actually conforms to the limiting law up to substantially higher concentrations: this is due to the effect of ion-pair formation, which, though negligible below 0.001 N, reduces the conductivity at higher concentrations by amounts comparable to those involved by neglecting the factor $(1 + \kappa a)$. Figure 7.1 illustrates the concordance between the experimental data of Shedlovsky for aqueous hydrochloric acid and the predictions of the Onsager

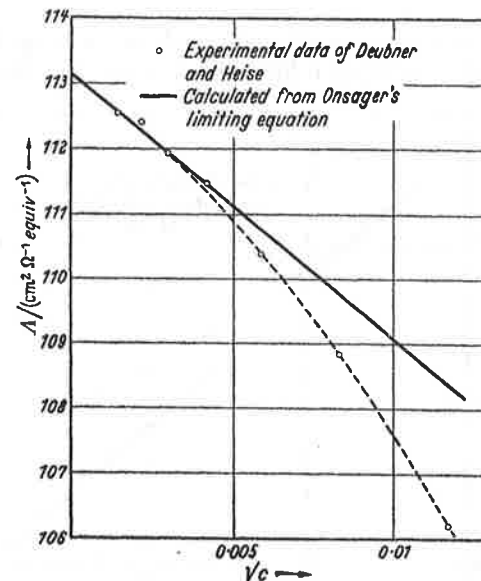


Figure 7.2. Conductivity of cadmium sulphate at 18°

limiting law at concentrations below $c = 0.003$. For 2 : 2 electrolytes and higher valency types, however, the limiting law is obeyed only at extraordinarily low concentrations, the formation of ion-pairs being appreciable even at high dilutions. It is only recently that evidence has been advanced to show that the limiting Onsager equation is valid for a 2 : 2 electrolyte. By taking extraordinary precautions, DEUBNER and HEISE⁽¹⁴⁾ have been able to measure the conductivity of cadmium sulphate solutions at concentrations as low as $c = 2 \times 10^{-6}$. Figure 7.2 shows how the seven values of the conductivity determined by Deubner and Heise do agree with the values predicted by the limiting law:

$$\Lambda = 113.15 - 408.1\sqrt{c}$$

It is clear that with increasing dilution the experimental values

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approach more nearly to the theoretical and are in good agreement for the four most dilute solutions. At higher concentrations the experimental values are lower than the predicted; as we shall see in Chapter 14 this is characteristic of salts subject to ion-pair formation. The deviation is in the opposite direction with non-associated salts, and Figure 7.3 illustrates this by using the data of Shedlovsky for sodium chloride up to comparatively high concentrations.

Valuable though the confirmation of the limiting law at these extreme dilutions is, there is little practical use for a theory dealing

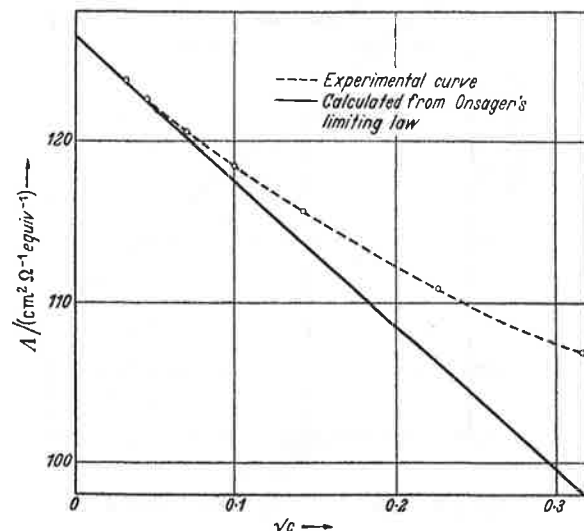


Figure 7.3. Equivalent conductivity of sodium chloride solutions at 25°

only with such solutions. More interest therefore attaches to testing the more complete equations for higher concentrations in which the factor $(1 + \kappa a)$ is not omitted; since these all necessarily reduce to the Onsager form when $\kappa a \ll 1$, it is clear that if they hold, the Onsager limiting law must also do so when only the concentration-range below about 0.001 N is considered.

The most careful experimental work has largely been confined to solutions less than 0.1 N in concentration, and the very precise direct current method of Gordon has been employed only up to 0.01 N. Below 0.01 N the best measurements by different observers often agree within about 0.03 in Λ , but at higher concentrations uncertainties of several tenths of a unit exist. For example, the

Table 7.3

Tests of Equation (7.36) $\Lambda^\circ = \Lambda + \frac{(B_1\Lambda + B_2)\sqrt{c}}{1 + (Ba - B_1)\sqrt{c}}$

Temp. °C	Λ° eq. (7.36)	Mean δ per cent	Max. δ per cent	Number of points	Range (molar)	Observer	Λ° (S.E.F.)
5°	297.61	0.03	0.09	12	0.001-0.083	O & S	297.6
15°	361.89	0.04	0.09	11	0.001-0.082	O & S	362.0
25°	425.98	0.03	0.06	12	0.002-0.086	O & S	426.2
25°	426.10	0.03	0.05	11	0.00003-0.003	S	426.16
35°	489.02	0.02	0.06	14	0.001-0.062	O & S	489.2
45°	550.18	0.02	0.05	11	0.002-0.090	O & S	550.3
55°	609.34	0.02	0.05	11	0.002-0.070	O & S	609.5
65°	666.64	0.02	0.06	12	0.001-0.072	O & S	666.8

O & S: OWEN, B. B. and SWEETON, F. H., *J. Amer. chem. Soc.*, 63 (1941) 2811

S: SHEDLOVSKY, T. (Λ° converted to Jones and Bradshaw 0.1 denal standard) *ibid.* 54 (1932) 1411

Λ° (S.E.F.) denotes the values obtained by the observers named, using the Shedlovsky extrapolation function (7.33)

From ROBINSON, R. A. and STOKES, R. H., *J. Amer. chem. Soc.*, 76 (1954) 1991 (where similar data are given for four 1:1 salts at various temperatures)

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equivalent conductivity of 0.1 N potassium bromide solution at 25° is given as 131.19 by JONES and BICKFORD⁽¹⁵⁾ and as 131.39 by LONGSWORTH⁽¹⁶⁾. However, since the significant quantity for the theory is ($\Lambda^0 - \Lambda$) and this is larger at higher concentrations, such discrepancies in the data are not serious; one should, however, be prepared to tolerate deviations of a few tenths of a unit in Λ at the higher concentrations.

All of the proposed equations can be expressed in the form:

$$\Lambda = \Lambda^0 + f(c, a)$$

so that the problem is to find the best values of the two constants Λ^0 and a . To illustrate the precision with which their equation will reproduce the experimental results for a strong 1 : 1 electrolyte, Fuoss and ONSAGER⁽⁹⁾ quote details for aqueous potassium bromide at 25°. The experimental data used are those of OWEN and ZELDES⁽¹⁷⁾ which cover the concentration range 0.0014–0.0072 N. By a method of successive approximations, the best values are found to be $\Lambda^0 = 151.75$, $a = 3.6 \text{ \AA}$, and the corresponding equation fits the data within 0.01 cm² ohm⁻¹ equiv.⁻¹.

Shedlovsky's empirical equation (7.34) will represent the same data with only slightly inferior accuracy but requires $\Lambda^0 = 151.68$. Our equation (7.36) gives a similar fit with $\Lambda^0 = 151.67$ and $a = 3.2 \text{ \AA}$. Equation (7.35), which is that of Falkenhagen's school, or that of Pitts without the 'higher terms,' requires an ion size of 2.0 Å and gives $\Lambda^0 = 151.71$. In Table 7.4 the predictions of these various equations are compared with the experimental results. Whilst the table shows that the Fuoss-Onsager treatment gives an almost perfect fit to this set of experimental results, it must be emphasized that the other theoretically less exact equations give deviations of at the worst only 0.02 cm² ohm⁻¹ equiv.⁻¹ or 0.014 per cent. The important question is therefore whether the Λ^0 value given by the Fuoss-Onsager theory is more correct than the others, which are 0.04 to 0.08 lower. There is no method of determining Λ^0 absolutely; it must always be found by extrapolation and the Fuoss-Onsager theory demands that the simpler extrapolation functions should curve upwards slightly in very dilute solutions. Whether this actually occurs is obscured by the increasing importance of experimental errors at high dilutions. Probably the most effective test can be made by using data for very dilute hydrochloric acid; here the solvent corrections are less important than for other electrolytes, and the data in the dilute region should be correspondingly more reliable. Application of the Fuoss-Onsager formulae to SHEDLOVSKY'S⁽¹⁸⁾ data for the range up to 0.003 N, gives $\Lambda^0 = 426.27$,

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compared with the values 426.16 obtained by Shedlovsky, and 426.10 using equation 7.36 with $a = 4.3 \text{ \AA}$, from the same data. It seems safe therefore to estimate that Λ^0 values obtained by the SHEDLOVSKY⁽¹¹⁾ or ROBINSON-STOKES⁽¹²⁾ methods from data in the range 0.001 – 0.01 N will not be more than 0.05 per cent lower than the Fuoss-Onsager values.

Bearing in mind that unless quite exceptional precautions are taken, the experimental data will not be of better than 0.02 per cent

Table 7.4
Equivalent conductance of potassium bromide solutions at 25°

	cm ² Int. Ω^{-1} equiv ⁻¹				
10 ⁴ c mole/l.	$\Lambda_{\text{exp.}}$	$\Lambda_{\text{F-O}}$	Λ_{S}	$\Lambda_{\text{R-S}}$	$\Lambda_{\text{P-F}}$
13.949	148.27	148.27	148.26	148.26	148.28
27.881	146.91	146.91	146.92	146.92	146.93
42.183	145.88	145.89	145.89	145.90	145.90
59.269	144.90	144.91	144.90	144.91	144.89
71.696	144.30	144.30	144.28	144.29	144.28

Λ^0	—	151.75	151.68	151.67	151.71
	—	$a = 3.6 \text{ \AA}$	$b = 91$	$a = 3.2 \text{ \AA}$	$a = 2.0 \text{ \AA}$

$\Lambda_{\text{F-O}}$ —calculated by Fuoss-Onsager theory including transcendental terms (ref. 9).

Λ_{S} —calculated by Shedlovsky's function, equation (7.34).

$\Lambda_{\text{R-S}}$ —calculated by Robinson-Stokes equation, (7.36).

$\Lambda_{\text{P-F}}$ —calculated by Falkenhagen or Pitt's equations, ignoring 'higher terms,' equation (7.35).

accuracy, it is clear that the simpler equations will be adequate for most work, and the arithmetical labour of applying the Fuoss-Onsager theory will be undertaken only when extreme precision is needed. To illustrate the reality of experimental errors, we compare in Fig. 7.4 the results for potassium bromide at 25° obtained by three investigators, OWEN and ZELDES⁽¹⁷⁾, BENSON and GORDON⁽¹⁹⁾, and JONES and BICKFORD⁽¹⁵⁾. The quantity plotted is the arbitrary deviation function ($\Lambda + 81\sqrt{c}$). It is evident that the differences between various investigators are as great as the differences between Λ^0 obtained by the various equations tested in Table 7.4.

Equation 7.36 is particularly convenient for representing conductances up to 0.1 N, though the best value of the parameter a and consequently of Λ^0 depend slightly on the concentration range fitted. Table 7.5 compares the experimental conductances of sodium chloride with the predictions of this equation using $a = 4 \text{ \AA}$; the

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deviations are less than 0.05 per cent up to 0.05 N. Since the equation involves little more calculation than does the limiting law (7.29),

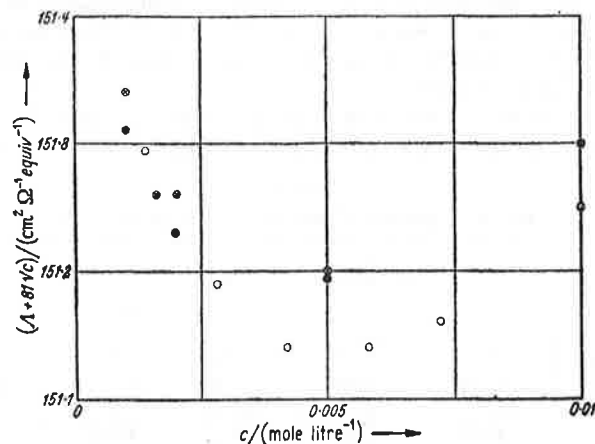


Figure 7.4. Deviation function for the conductivity of aqueous potassium bromide solutions at 25° according to various investigators

○ Owen and Zeldes. ● Benson and Gordon. ⊗ Jones and Bickford

its slight theoretical inadequacy compared with the more exact equation of Fuoss and Onsager may well be forgiven.

Table 7.5
Conductivities of Sodium Chloride Solutions at 25°

c mole/l.	Λ_{obs}	Λ (Eq. 7.36)	$\Lambda_{\text{L.L.}}$ (Eq. 7.29)
0	(126.45)	(126.45)	(126.45)
0.0005	124.51	124.51	124.45
0.001	123.74	123.75	123.63
0.002	122.66	122.68	122.46
0.005	120.64	120.68	120.14
0.01	118.53	118.57	117.53
0.02	115.76	115.81	113.83
0.05	111.06	111.03	106.50
0.1	106.74	106.52	98.23

Λ (Eq. 7.36) calculated with $a = 4\text{\AA}$;
 $\Lambda_{\text{L.L.}}$ by Onsager limiting law.

LIMITATIONS OF CONDUCTIVITY EQUATIONS

In deriving the various equations for the conductivity discussed above, the following assumptions are made, each resulting in some restriction on the applicability of the final equations. (a) Complete

VARIATION OF TRANSPORT NUMBERS WITH CONCENTRATION

ionization is assumed; the formulae will however apply to the ionized part of weak electrolytes and to the 'non-paired' part of electrolytes in which ion-association occurs. In practice, only a small number of 1 : 1 electrolytes in water and possibly in some other solvents of high dielectric constant can be treated as completely ionized. Fuoss and Onsager comment that the mere fact of introducing the ion size parameter a implies some degree of ion-association, since it involves the idea that some ions do approach to mutual contact and whilst in mutual contact they will not contribute to the conductance. (This appears to us to be not quite the case, for even whilst the ions are in contact they can contribute something to the conductance by moving round each other.) (b) The treatments of Fuoss and Onsager and of Falkenhagen are based on the Debye-Hückel expression (4.13) for the potential in the absence of an external field. The limitations of this have been discussed in Chapter 4; it is an approximation which accords well with thermodynamic data (see Chapter 9) and is most accurate for ions of low charge in media of high dielectric constant. The more complicated potential expression employed by Pitts and by Mirskhulava is a doubtful improvement in view of the departure from mathematical self-consistency involved.

The theory is weaker at every point for unsymmetrical electrolytes: the potential expression is less exact since the term in ψ^2 does not vanish from equation (4.7); the convergence for the series for the electrophoretic effect is unsatisfactory; and the theory of the relaxation effect has not been properly developed beyond the first approximation given by equation (7.9). There is therefore little justification for using any theoretical treatment except the Onsager limiting law (7.29) in such cases; empirical terms in c , $c \ln c$, $c^{3/2}$ etc., may be added but only for convenience of representation. Mere division of $\sqrt{I}$ in equation 7.29 by $(1 + \kappa a)$ does indeed give a rough fit, and with reasonable ion size parameters, but the discrepancies between its predictions and the measurements are much greater than experimental error. Two examples are given in Table 7.6.

Even for aqueous 1 : 1 electrolytes, one is straining the mathematics to the limit in applying theory to solutions as strong as 0.1 N, and only approximate treatments can be given at higher concentrations. Some of these will be discussed in Chapter 11.

THE VARIATION OF TRANSPORT NUMBERS WITH CONCENTRATION

Experimental results show that transport numbers are in general concentration-dependent, and the interpretation of this observation

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provides a useful test of the theory. In the case of non-associated uni-univalent electrolytes the form of the concentration dependence is as follows:

- (a) If the cation transport number is close to 0.5, it scarcely varies with concentration; this is seen in the case of potassium chloride.

Table 7.6

Conductivity of Calcium and Lanthanum Chloride Solutions at 25°

CaCl ₂ (i)				LaCl ₃ (ii)			
<i>c</i>	Λ _{obs.}	Λ _{calc.}	Λ _{L.L.}	<i>c</i>	Λ _{obs.}	Λ _{calc.}	Λ _{L.L.}
0	(135.85)	—	—	0	(145.9)	—	—
0.00025	131.90	132.02	131.88	0.000167	139.6	139.9	139.6
0.0005	130.32	130.52	130.23	0.000333	137.0	137.6	137.0
0.0010	128.20	128.47	127.91	0.00167	127.5	128.8	126.0
0.0015	126.61	126.95	126.12	0.00333	121.8	123.0	117.8
0.0025	124.23	124.65	123.29	0.00667	115.3	115.9	106.2
0.0035	122.47	122.85	120.99	0.0167	106.2	104.3	83.1
0.0050	120.36	120.69	118.09	0.0333	99.1	94.3	57.1
0.01	115.65	115.65	110.73				
0.025	108.47	107.19	96.14				
0.05	102.46	99.52	79.68				

(*a* = 4.31 Å)

(*a* = 4.9 Å)

- (i) Data up to *c* = 0.005 from BENSON, G. C. and GORDON, A. R., *J. chem. Phys.*, 13 (1945) 470; above *c* = 0.005 from SHEDLOVSKY, T. and BROWN, A. S., *J. Amer. chem. Soc.*, 56 (1934) 1066
 (ii) JONES, G. and BICKFORD, C. F., *J. Amer. chem. Soc.*, 56 (1934) 602; LONGSWORTH, L. G. and MACINNES, D. A., *ibid.*, 60 (1938) 3070

- (b) If the cation transport number is less than 0.5 as with lithium chloride, it decreases further with increasing concentration.
 (c) If the cation transport number is greater than 0.5, it increases with concentration; this occurs for example with hydrochloric acid.

These findings are completely and quantitatively explained by the interionic attraction theory⁽²⁰⁾. According to equations (7.25), (7.27) and (7.36) the transport number *t*₁ of the cation is given by:

$$t_1 = \frac{\lambda_1}{\Lambda} = \frac{\lambda_1^0 - \frac{1}{2}|z_1|B_2\sqrt{I}/(1 + \kappa a)}{\Lambda^0 - \frac{1}{2}(|z_1| + |z_2|)B_2\sqrt{I}/(1 + \kappa a)} \dots (7.39)$$

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where

$$B_2 = \frac{82.5}{\eta(\epsilon T)^{1/2}}, \text{ and } \kappa a = Ba\sqrt{I}$$

For 1 : 1 electrolytes this simplifies to:

$$t_1 = \frac{\lambda_1^0 - \frac{1}{2}B_2\sqrt{c}/(1 + \kappa a)}{\Lambda^0 - B_2\sqrt{c}/(1 + \kappa a)} \dots (7.40)$$

Thus the transport number expression contains only electrophoretic terms, the relaxation factor having cancelled out from equations (7.25) and (7.27). It is clear from (7.39) that if the limiting transport number $t_1^0 = \frac{\lambda_1^0}{\Lambda^0}$ is 0.5 exactly, *t*₁ will not vary from this value, and

that the behaviour described under (b) and (c) above is also accounted for. Equation (7.39) gives an excellent quantitative account of the observed transport numbers; for aqueous 1 : 1 electrolytes at 25°, *B*₂ = 60.65 and *κa* = 0.3291 *a*√*c*. Table 7.7 gives some observed and calculated values of *t*₁. The high degree of agreement with theory exhibited in Table 7.7 is striking evidence for the soundness of the treatment of the electrophoretic effect for 1 : 1 electrolytes. The values of the ion size parameter *a* needed to account for the transport numbers are very reasonable, and similar to those found from consideration of the activity coefficient data. The ions of potassium chloride, for example, appear from their mobilities to be very nearly of the same effective size, and the crystal radius of the chloride ion is 1.8 Å; the value *a* = 3.7 Å is thus just about what would be expected. For the fairly strongly hydrated sodium and lithium ions we have estimated in Chapter 6 radii of 3.3 and 3.7 Å respectively by using a modified Stokes' law formula: combining these with 1.8 Å for the chloride ion, we have *a* = 5.1 and 5.5 Å for sodium chloride and lithium chloride respectively, as compared with the value *a* = 5.2 Å found adequate for both salts in Table 7.7. A Stokes' law estimate cannot be used for hydrogen ion because of the abnormal transport mechanism involved, but the value *a* = 4.4 Å for hydrochloric acid compares very well with *a* = 4.47 Å required for the activity coefficient data (Chapter 9).

Very few transport number data are available for solvents other than water, and such as there are have mostly been obtained in mixed solvents, e.g., the transport numbers of hydrochloric acid have been measured by HARNED and DREBY⁽²¹⁾ in a number of dioxane-water mixtures, and GORDON and his collaborators⁽²²⁾ have used the moving boundary method for sodium and potassium chlorides in equimolar methanol-water mixtures. These results for

Table 7.7
Observed and Calculated Cation Transport Numbers of Aqueous 1:1 Electrolytes at 25°: Tests of Equation (7.39)

	HCl*†		LiCl*‡		NaCl§		NaAc‡		KCl*§		KAc	
	t_1 obs.	t_1 calc.	t_1 obs.	t_1 calc.	t_1 obs.	t_1 calc.	t_1 obs.	t_1 calc.	t_1 obs.	t_1 calc.	t_1 obs.	t_1 calc.
0	(0.8709)	0.8209	(0.3363)	0.3363	(0.3962)	0.3962	(0.5506)	0.5506	(0.4905)	0.4905	(0.6425)	0.6425
0.01	0.8251	0.8249	0.3289	0.3285	0.3918	0.3918	0.5537	0.5538	0.4902	0.4901	0.6498	0.6495
0.02	0.8266	0.8263	0.3261	0.3258	0.3902	0.3902	0.5550	0.5550	0.4901	0.4900	0.6523	0.6521
0.05	0.8292	0.8287	0.3211	0.3211	0.3876	0.3875	0.5573	0.5573	0.4899	0.4898	0.6569	0.6570
0.1	0.8314	0.8310	0.3168	0.3165	0.3854	0.3849	0.5594	0.5596	0.4898	0.4895	0.6609	0.6619
0.2	0.8337	0.8337	0.3112	0.3112	0.3821	0.3819	0.5610	0.5626	0.4894	0.4892	—	—
0.5	0.838	0.838	0.303	0.301	—	—	—	—	0.4888	0.4887	—	—
1.0	0.841	0.841	0.297	0.287	—	—	—	—	—	—	—	—
2.0	0.843	0.843	—	—	—	—	—	—	0.4882	0.4883	—	—
3.0	0.843	0.845	—	—	—	—	—	—	—	—	—	—
a (Å)	4.4			5.2		3.7		3.7		3.7		3.7

* LONGSWORTH, L. G., *J. Amer. chem. Soc.*, 54 (1932) 2741.

† HANSEN, H. S. and DEGEN, E. C., *ibid.*, 61 (1939) 3113.

‡ LONGSWORTH, L. G., *ibid.*, 57 (1935) 1185.

§ ALLCOCK, R. W., LE ROY, D. J. and GORDON, A. R., *J. chem. Phys.*, 8 (1940) 418 and 10 (1942) 124.

|| LE ROY, D. J. and GORDON, A. R., *ibid.*, 6 (1938) 398.

Note: Experimental values given to four figures are by moving boundary method; those given to three figures are by the e.m.f. method for hydrochloric acid and by the Hittorf method for lithium chloride.

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hydrochloric acid appear to conform reasonably well with the requirements of the theory at low concentrations, but not so well at higher concentrations. This is perhaps to be expected since the convergence of the theoretical formulae for the electrophoretic effect is less satisfactory in media of low dielectric constant.

For higher valency types, even in water, the theory is also found to be inadequate; with calcium chloride, for example, equation (7.39) gives results which are nearer to the observed values than are those predicted by the limiting law, but are still substantially low. In 0.05 M calcium chloride at 25° the observed cation transport number is 0.4070; equation (7.39) with $a = 5$ Å gives 0.3952; and the limiting law equation (*i.e.*, equation (7.39) with $a = 0$) gives 0.3545. Thus while the ion-size correction gives a useful improvement, it does not lead to the quantitative agreement which can be obtained with 1:1 electrolytes. This is probably attributable to the lower degree of self-consistency of the interionic attraction theory for unsymmetrical electrolytes.

For bi-bivalent electrolytes (*i.e.*, of the zinc sulphate type), the theory should be applicable; but here the difficulty arises that a large proportion of the ions are present as closely associated ion-pairs; this effect is quite important at the lowest concentration accessible to transport number experiments, *viz.*, about 0.005 M. Unfortunately no moving-boundary measurements have been made on 2:2 electrolytes, and we therefore have to rely on older and less precise measurements by the Hittorf method. The measurements on cadmium sulphate at 18° by JAHN and his co-workers⁽²³⁾ seem to be the best available, and internal evidence suggests a reliability of a few units in the third decimal place of the transport number. The curve of t_1 against $\sqrt{c}$ also bears a strong resemblance to that found by the electromotive force method⁽²⁴⁾ for zinc sulphate, except in the most dilute region, where the Hittorf method may well be more reliable.

The transport number of the cadmium ion is found to fall almost linearly in $\sqrt{c}$ from $t_1^0 = 0.396$ at $c = 0$ to $t_1 = 0.254$ at $c = 1$ mole per litre; deviations from this straight line scarcely exceed the experimental error. Taking the limiting equivalent conductivities at 18° as $\lambda_{Cd^{++}}^0 = 44.8$, $\lambda_{SO_4^{--}}^0 = 68.4$, and inserting the appropriate numerical values for 18° in equation (7.39), we have:

$$t_1 = \frac{44.8 - 101.7\sqrt{c}/(1 + 0.6546 \times 10^8 a\sqrt{c})}{113.2 - 203.4\sqrt{c}/(1 + 0.6546 \times 10^8 a\sqrt{c})} \dots (7.41)$$

With the value $a = 3.5$ Å, equation (7.41) gives very fair agreement with experiment as shown in Table 7.8. The formation of

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ion-pairs in a symmetrical salt does not result in the appearance of any new ionic species and the effect on the transport number should therefore be merely that of 'diluting' the solution by removing some of the ions to form electrically neutral pairs, the 'dilution'

Table 7.8
Cation Transport Number in Cadmium Sulphate at 18°

c (mole/litre)	0	0.01	0.09	0.25	0.49	1.00
t_1 obs.	(0.396)	0.384	0.353	0.323	0.295	0.254
t_1 calc.	0.396	0.378	0.347	0.321	0.299	0.270

factor being α , the degree of dissociation of the ion-pairs. The success of equation (7.41) in which the ion pairing is ignored is apparently due to this 'dilution factor' being approximately compensated by the use of a rather small value (3.5 Å) for the ion-size parameter.

Negative cation transport numbers

The behaviour of the transport numbers of calcium chloride or zinc perchlorate may be regarded as normal for 2:1 electrolytes in spite of the inability of theory to cope with it. Many of the

Table 7.9
Cation Transport Numbers of Aqueous 2:1 Electrolytes at 25°, showing Effect of Autocomplex Formation in Zinc Halides

m	$\text{Zn}(\text{ClO}_4)_2^*$	$\text{ZnI}_2^\dagger$	$\text{ZnBr}_2^\ddagger$	$\text{ZnCl}_2^\S$
0	(0.440)	(0.408 ₆)	(0.404 ₁)	(0.409 ₇)
0.05	—	0.382	0.366	0.365
0.1	0.409	0.363	0.349	0.350
0.2	0.389	0.345	0.331	0.335
0.5	0.361	0.320	0.306	0.331
1.0	0.335	0.291	0.286	0.171
2.0	0.303	0.178	0.181	0.000
3.0	0.281	0.056	— 0.059	— 0.137
4.0	0.271	— 0.050	— 0.151	— 0.256
5.0	—	— 0.190	— 0.233	— 0.364
8.0	—	— 0.444	— 0.445	— 0.562
10.0	—	— 0.550	— 0.563	— 0.559

m = mole salt per kg water

* STOKES, R. H. and LEVIEN, B. J., *J. Amer. chem. Soc.*, 68 (1946) 333

† STOKES, R. H. and LEVIEN, B. J., *ibid.*, 68 (1946) 1852

‡ PARTON, H. N. and MITCHELL, J. W., *Trans. Faraday Soc.*, 35 (1939) 758

§ HARRIS, A. C. and PARTON, H. N., *ibid.*, 36 (1940) 1139

transition-metal halides⁽²⁵⁾, however, show very different behaviour, which is illustrated in Table 7.9; at high concentrations the cation

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transport number decreases rapidly to zero, then becomes negative. This is in marked contrast to the behaviour of the transport number of zinc in its perchlorate⁽²⁶⁾ which may be taken as typical behaviour for a normal 2:1 electrolyte at higher concentrations; the explanation of the anomaly is that the metal ion is to a large extent present as a complex negative ion, believed to be mainly an ion of type ZnX_4^- , a view which is supported by measurements⁽²⁷⁾ of the vapour pressures of ZnX_2 - KX mixtures. Though no quantitative treatment of the effect can be given, it would appear (p. 112) that to give a negative apparent transport number for the metal ion, the complex negative ion must have a higher mobility than the normal (hydrated) metal ion. This is quite possible since the bivalent cations are known to be strongly hydrated.

CONDUCTIVITIES IN NON-AQUEOUS SOLVENTS

There is a substantial body of experimental data for conductivities in non-aqueous solvents. Such solvents frequently have a much lower conductivity than pure water, with the result that measurements can be made at lower concentrations without serious loss of accuracy; on the other hand they are more difficult to purify, and may require careful protection from atmospheric moisture, while simple salts are often only slightly soluble in them, with a consequent restriction of the concentration-range which can be studied. The theoretical interpretation of the results is at present hampered by a severe shortage of reliable transport numbers in non-aqueous solvents. It is to be hoped that the centrifugal cell method of MacInnes will soon be developed to a point where this difficulty is overcome; in the meantime a valuable start has been made by Gordon and his collaborators who have obtained accurate moving-boundary transport number measurements in anhydrous methanol solutions of sodium and potassium chlorides. These, together with their direct-current conductivity measurements in the same solvent⁽²⁸⁾, provide the most precise information we have on the transport properties of ions in non-aqueous solutions. The results of their measurements are summarized in Table 7.10. Both cation and anion transport numbers were measured in several cases, the sum being within 0.0003 of unity; this provides a valuable check on the results.

GORDON's school⁽²⁹⁾ have also made measurement of transport numbers and conductances for lithium, sodium and potassium chlorides in anhydrous ethanol. Owing to the low solubilities, (the maximum concentration used was 0.0025 N), and to the occurrence

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of ion-association, only their limiting ion conductivities are given in Table 7.10.

First, examining the limiting ionic conductivities, we see that for both anions and cations the λ^0 values increase as the crystal radii

Table 7.10
Transport Numbers in Methanol at 25°

c	t_1 (NaCl)	t_1 (KCl)
0	(0.4633)	(0.5001)
0.003	0.4603	—
0.005	0.4595	0.5007
0.007	0.4588	0.5009
0.01	0.4582	0.5013
0.02	—	0.5012

c in mole/litre

From DAVIES, J. A., KAY, R. L. and GORDON, A. R., *J. chem. Phys.*, 19 (1951) 749

Equivalent Conductivities, Λ , in Methanol at 25°

$c \times 10^4$ mole/l	LiCl	NaCl	NaBr	KCl	KBr	KI
0	(92.20)	(97.61)	(101.76)	(104.78)	(108.95)	(115.15)
1	89.74	—	99.19	—	106.34	112.52
2	88.70	94.11	98.11	101.16	105.26	111.43
5	86.65	92.09	96.04	99.07	103.10	109.29
10	84.52	89.87	93.80	96.72	100.71	106.94
20	81.74	86.91	90.86	93.56	97.51	103.74
30	79.73	84.84	88.80	91.24	95.19	101.50
50	76.73	81.80	85.66	87.79	91.80	98.16
70	—	79.43	—	85.28	—	—
100	—	76.71	—	82.32	—	—

BUTLER, J. P., SCHIFF, H. I. and GORDON, A. R., *J. chem. Phys.*, 19 (1951) 752;
JERVIS, R. E., MUIR, D. R., BUTLER, J. P. and GORDON, A. R., *J. Amer. chem. Soc.*, 75 (1953) 2855

Limiting Ionic Conductivities in Methanol and Ethanol at 25°

Ion	Li ⁺	Na ⁺	K ⁺	Cl ⁻	Br ⁻	I ⁻
$\lambda^0(\text{MeOH})$	39.82	45.22	52.40	52.38	56.55	62.75
$\lambda^0(\text{EtOH})$	17.05	20.31	23.55	21.85	—	—
$r(\text{\AA})$	0.60	0.95	1.33	1.81	1.95	2.16

increase, though the anion and cation values do not fall on the same curve. This is a slightly more regular situation than prevails for the same ions in water, where for anions the order of increasing

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mobilities bears no relation to the crystal radii. It is unlikely that any of the anions carries a permanent solvation sheath in methanol, but the increase in mobility with size may be due to decreased interaction of the larger ion with the solvent dipoles. Since the molal volume of methanol is about 41 cm³, the 'radius' of a methanol molecule must be substantially greater than that of a water molecule, so that the conditions for the validity of Stokes' law are far from realization with these small ions. One may perhaps picture the ion as causing some rotation of solvent dipoles as the ion passes the molecule, which will result in a dissipation of energy and will therefore increase the effective resistance; this interaction will increase rapidly with reduction of the ion-to-dipole distance, as required by the observations. Such an effect might well be more important in methanol than in water, where the 'structure' of the solvent is more definite. The possible existence of this effect requires further investigation.

The variation with concentration of the transport number of potassium chloride in methanol is scarcely significant, as is required by the theory for cases where the limiting transport numbers are nearly 0.5. That of sodium chloride may profitably be examined by the theory. For methanol at 25°, the viscosity is 0.005445 poise and the dielectric constant 31.52*; these give for the values of the constants in equation (7.36):

$B = 0.5188 \times 10^8$, $B_1 = 0.9004$, $B_2 = 156.2$; or in the transport number equation (7.39), the constant $B_2 = 156.2$ and $\kappa a = 0.5188 \times 10^8 a \sqrt{c}$. The equation:

$$t_1 = \frac{45.22 - 78.1 \sqrt{c}/(1 + 3.06 \sqrt{c})}{97.61 - 156.2 \sqrt{c}/(1 + 3.06 \sqrt{c})}$$

reproduces the observed transport numbers up to 0.01 N within 0.0001. The ion size corresponding to $\kappa a = 3.06 \sqrt{c}$ is $a = 5.9 \text{ \AA}$, a rather large value unless the sodium ion at least is solvated, but about the value one would expect if the ions approach until separated by one methanol molecule only.

In this medium, of dielectric constant 31.52, a considerable amount of ion-pair formation is to be expected even with 1:1 electrolytes, since the Bjerrum critical distance is 8.9 Å. We have remarked before that ion-pair formation in a symmetrical electrolyte will affect the transport numbers only by a sort of 'dilution' effect. The equivalent conductivities on the other hand, will be reduced in nearly direct proportion to the amount of ion-pairing; they are much more sensitive to it than are transport numbers. It is therefore

* A more recent value is 32.63.

not surprising to find that the conductivities of the alkali halides in methanol conform rather too closely to the Onsager limiting law. For sodium chloride, for example, the limiting law becomes:

$$\Lambda = 97.61 - 244.1\sqrt{c}$$

which at 0.001 N gives $\Lambda = 89.89$ in agreement with the observed value $\Lambda = 89.87$. Since the value of (κa) must be at least 0.05, this agreement is too close for a fully ionized electrolyte. In general, the conductivities at higher concentrations lie above the limiting law values, but not as much as they should for fully dissociated electrolytes. The values cannot be accurately represented by equation (7.36); a value of $a = 3.2 \text{ \AA}$ for sodium chloride gives a rough fit, within about 0.5 in Λ , but this value of the ion size is inconsistent with that needed for the transport number equation.

Liquid hydrogen cyanide is of great interest as an electrolytic solvent, having a dielectric constant of about 160 at 0° and 120 at 18°, so that ion-pair formation should be less than in water. It has also a much lower viscosity than that of water, the values at 0° being 0.00232 poise for hydrogen cyanide and 0.01787 poise for water. COATES and TAYLOR⁽³⁰⁾ studied a number of alkali-metal salts in this solvent at 18°, and LANGE, BERGÅ and KONOPK⁽³¹⁾ have made measurements at 0° on some potassium and some tetra-substituted ammonium salts. The 18° measurements were all at low concentrations (0.0001–0.0025 mole per litre) and conform to relations of the type:

$$\Lambda = \Lambda^0 - A\sqrt{c}$$

with Λ^0 and A values given in Table 7.11. These linear relations hold over the whole of the concentration-range studied (up to 0.002 or 0.003 N in most cases) except for lithium chloride, nitrate and thiocyanate and sodium nitrate. These show a downward curvature, most marked in the case of lithium thiocyanate, and probably indicative of ion-pair formation. The limiting conductivities are reasonably consistent with the Kohlrausch principle, as shown by the nearly constant differences:

$$\Lambda_{\text{LiX}}^0 - \Lambda_{\text{NaX}}^0 \approx 3$$

$$\Lambda_{\text{KX}}^0 - \Lambda_{\text{NaX}}^0 \approx 19.6$$

The range of Λ^0 values is noticeably more restricted than in water, and there are indications that there is less actual solvation of ions than occurs in water. The values of the slopes (A) of the Λ versus $\sqrt{c}$ curves do not agree any too well with the theoretical limiting result, which for hydrogen cyanide at 18° becomes:

$$\Lambda = \Lambda^0 - [0.1271\Lambda^0 + 233]\sqrt{c}$$

Since the electrophoretic term is here much larger than the relaxation term, all the theoretical slopes are much the same, lying in the range $A = 259$ –269. In the cases where the curvature of the plots suggests ion-pair formation, the observed slope in the most dilute

Table 7.11

Conductivities of Salts in Hydrogen Cyanide Solution at 18°

$$\Lambda = \Lambda^0 - A\sqrt{c}$$

	Λ^0	A		Λ^0	A
LiCl	345.4	335	Na Picrate	266.9	195
LiBr	346.9	270	KCl	363.4	280
LiI	348.0	258	KBr	363.2	248
LiNO ₃	336.6	402	KI	363.9	235
LiClO ₄	336.9	230	KNO ₃	353.9	253
LiCNS	340.6	400	KClO ₄	353.3	275
NaBr	343.8	243	KCNS	358.0	243
NaI	344.9	238	RbCl	363.2	195
NaNO ₃	333.8	250	CsCl	368.2	200
NaClO ₄	335.5	235	N(Et) ₄ Picrate	282.3	215
NaCNS	337.7	230			

The A values given are the experimentally observed slopes; theoretical values of A lie between 259 and 269.

COATES, J. E. and TAYLOR, E. G., *J. chem. Soc.* (1936) 1245

region is considerably steeper than the theoretical, which is reasonable; but the remaining presumably 'normal' salts appear to give straight lines lying above the theoretical slope. This is most noticeable with the picrates and rubidium and caesium chlorides, which have large ions. For the salts such as sodium bromide, where the points lie only slightly above the limiting-law lines, the introduction of the factor $(1 + \kappa a)$ into the denominator, as required by the more complete theory, gives a satisfactory account of the results. In this solvent at 18°, $\kappa = 0.2703 \times 10^8 \sqrt{c}$ and for sodium bromide the value $a = 5.1 \text{ \AA}$ in the equation:

$$\Lambda = \Lambda^0 - \frac{(0.1271\Lambda^0 + 223)}{(1 + \kappa a)}\sqrt{c}$$

is adequate and reasonable. Caesium chloride, however, requires $a = 28 \text{ \AA}$, which is quite absurd in view of the not greatly different limiting conductivities, which imply ions of comparable size to those of sodium bromide, but the concentrations are rather too low to permit an accurate evaluation of the parameter a .

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The measurements at 0° in hydrogen cyanide are rather more interesting, as they extend to concentrations high enough to make a significant test of equation (7.36) involving the ion-size correction. Data for potassium iodide solutions⁽³¹⁾, interpolated to round concentrations, are given in Table 7.12. For this solvent at 0°C, the viscosity (η) is 0.00232 poise, and the dielectric constant (ϵ) is 161. (The latter figure is by no means well established, however.) These values give the following values of the constants in equation (7.36):

$$B_1 = 0.0890, B_2 = 169.6, B = 0.240 \times 10^8$$

Now if the ions of potassium iodide are unsolvated in hydrogen cyanide solution, the distance of closest approach may be estimated

Table 7.12
Conductivities of Potassium Iodide in
Hydrogen Cyanide at 0°

c	$\Lambda_{\text{obs.}}$	$\Lambda_{\text{calc.}}$
0	(310.3)	(310.3)
0.001	304.4	304.2
0.002	301.8	301.8
0.003	300.2	300.0
0.005	297.3	297.1
0.007	294.9	294.9
0.010	292.1	292.1
0.015	288.1	288.4
0.02	284.9	285.4
0.05	269.6	273.2
0.10	252.2	261.0

c in mole/litre

$a = 3.5 \text{ \AA}$

Data from LANGE, J., BERGÅ, J. and KONOPIK, N., *Monatsh.*, 80 (1949) 708

from the crystal radii as $a \approx 1.33 + 2.16 \approx 3.5 \text{ \AA}$. Taking the limiting value of the conductivity as $\Lambda^0 = 310.3$, we then have for this solution from equation (7.35):

$$\Lambda_{\text{calc.}} = 310.3 - \frac{197.2\sqrt{c}}{1 + 0.84\sqrt{c}}$$

The values of $\Lambda_{\text{calc.}}$ given by this equation are included in Table 7.12; the agreement is quantitative up to 0.01 N and satisfactory up to 0.02 N, after which the calculated values are increasingly high.

On the whole, the conductivity measurements in hydrogen

Table 7.13
Limiting Conductances in Amide Solvents, Λ^0 in $\text{cm}^2 \Omega^{-1} \text{equiv}^{-1}$; η in poise

Solvent	Temp. °C	η	ϵ	HI	NaI	KI	CsI	Me ₄ NI	Et ₄ NI	Bu ₄ NI
Formamide (32)	25	0.0330	109.5	27.4	26.8	29.3	—	—	—	23.5
N-methyl formamide (33)	25	0.0165	182.4	—	44.4	45.0	47.2	—	49.0	—
N,N-dimethyl formamide (33, 34)	25	0.00796	36.7	—	82.0	82.6	—	91.0	87.5	77.7
N-methyl acetamide (35, 36)	40	0.0302	165.5	23.7	22.8	23.0	(LiI (21.2))	26.6	26.2	22.4
N,N-dimethyl acetamide (37)	25	0.00919	37.8	—	67.6	67.1	—	—	74.5	64.6
N-methyl propionamide (30°-60°) (38)	30	0.0457	164.3	—	13.4	13.7	—	—	—	—
N-methyl butyramide (30°-60°) (38)	30	0.0747	124.7	—	(NaCl 6.3)	—	—	—	—	—

Table 7.13—Contd.

Solvent	Temp. °C	Me ₃ PhNI	KCl	KBr	KPi (picrate)	KSCN	KNO ₃	KPhSO ₃	KClO ₄
Formamide (32)	25	27.3	29.8	—	—	—	—	23.1	—
N-methyl form- amide (33)	25	—	41.9	43.7	35.3	—	—	—	—
N-N-dimethyl form- amide (33, 34)	25	—	—	84.1	—	90.3	88.1	—	82.8
N-methyl acetamide (35, 36)	40	24.8	19.9	21.2	20.2	24.5	22.9	18.7	25.2
N-N-dimethyl acetamide (37)	25	70.1	—	68.5	56.8	74.1	71.6	56.3	68.1
N-methyl propion- amide (30°–50°) (38)	30	—	11.6	12.4	—	—	—	—	—
N-methyl butyramide (30°–50°) (38)	30	—	6.5	—	—	—	—	—	—

cyanide thus tend to support the theory; but there are some anomalies which clearly call for further investigation. Determinations of transport numbers, verification of the dielectric constant, and studies of the effect of solutes on the viscosity would all be of great value.

The amides of the lower aliphatic acids and their N-methyl derivatives form a class of liquids with extremely high dielectric constants (Appendix 1.2), the most striking case being that of N-methyl formamide $\text{H.CO.NH(CH}_3\text{)}$ which has $\epsilon = 182.4$ at 25°. Comprehensive studies of conductances in these solvents have recently been made by Sears and Dawson and co-workers and by French and Glover. In the case of formamide, an approximate measurement of the limiting transport number by the Hittorf method has also been made, giving $t_{\text{K}^+}^{\text{KCl}} = 0.406$ at 25°, so that individual ionic mobilities in this solvent are known. In the other solvents of this class no measured transport numbers are yet available, but reasonable estimates have been made on the basis of the behaviour of very large ions in relation to solvent viscosity. The original data are too extensive to present in detail, but Table 7.13 gives a summary of the major results. For compactness we have given Λ^0 values only of iodide ion with various cations and of potassium ion with various anions; these do not necessarily represent salts actually studied in the original work, but in some cases have been obtained by the application of the Kohlrausch principle to measurements on related salts.

A noteworthy feature is that the Λ^0 values for strong acids are similar to those for salts; evidently the hydrogen ion has no special transport mechanism available in these solvents, as it has in water and the lower alcohols.

The general pattern of concentration-dependence of the conductance is, as might be expected, one of approach to the Onsager limiting law from above; but in the di-N-methyl amides, which have dielectric constants about half that of water, the results lie close to the limiting-law curve, and with some salts actually fall below it, indicating a slight degree of ion-association.

Most of the other commonly used non-aqueous solvents have lower dielectric constants than methanol, and the conductivity of solutions in these appears to be so strongly influenced by ion association that little progress can be made by attempting to treat them as strong electrolytes. The extensive researches of Kraus and his collaborators have done much towards elucidating the behaviour of ion-aggregates in these solutions, and are discussed more fully in Chapter 14.

7 VARIATION OF CONDUCTIVITIES WITH CONCENTRATION

APPENDIX TO THE THEORY OF THE ELECTROPHORETIC EFFECT

EVALUATION OF THE INTEGRAL $S_n(\kappa a)$ OF EQUATION (7.5)
This integral takes an elementary form only for the case $n = 1$, when it becomes:

$$S_1(\kappa a) = \frac{1}{a} \int_a^\infty e^{-\kappa r} dr = \frac{e^{-\kappa a}}{\kappa a}$$

For $n \geq 2$ it involves the exponential integral function $Ei(x)$ defined by:

$$Ei(x) = \int_x^\infty e^{-y} y^{-1} dy$$

(y being merely a variable of integration). This function is available from tables⁽⁸⁹⁾ for various values of x .

For $n = 2$

$$S_2(\kappa a) = \int_a^\infty e^{-2\kappa r} r^{-1} dr = \int_a^\infty e^{-2\kappa r} (2\kappa r)^{-1} d(2\kappa r) = Ei(2\kappa a)$$

For $n > 2$ it is necessary to perform successive integrations by parts until the integral reduces to an exponential integral function.

Thus one obtains:

$$S_3(\kappa a) = e^{-3\kappa a} - 3\kappa a Ei(3\kappa a)$$

$$S_4(\kappa a) = e^{-4\kappa a} \left(\frac{1}{2} - 2\kappa a \right) + 8(\kappa a)^2 Ei(4\kappa a)$$

and in general for $n > 2$:

$$S_n(\kappa a) = e^{-n\kappa a} \left[\frac{1}{n-2} + \frac{(-n\kappa a)}{(n-2)(n-3)} + \frac{(-n\kappa a)^2}{(n-2)(n-3)(n-4)} + \dots + \frac{(-n\kappa a)^{n-3}}{(n-2)!} \right] + \frac{(-n\kappa a)^{n-2}}{(n-2)!} Ei(n\kappa a)$$

there being $(n-2)$ terms in the series enclosed in the square bracket. If the theory is taken only as far as the 'self-consistent' approximation, only $S_1(\kappa a)$ and $S_2(\kappa a)$ are involved, and the latter appears only in the theory of diffusion for symmetrical electrolytes. The higher-order terms have been computed to facilitate the investigation of questions of convergence.

EQUATIONS FOR CONDUCTIVITY AND TRANSPORT NUMBERS

SUMMARY OF EQUATIONS FOR CONDUCTIVITY AND TRANSPORT NUMBERS

1 = cation, 2 = anion

$$q = \frac{|z_1 z_2|}{(|z_1| + |z_2|)(|z_1| t_1^0 + |z_2| t_2^0)} = \frac{1}{2} \text{ for symmetrical valence-types} \quad \dots (7.10)$$

$$\kappa^2 = \frac{4\pi e^2}{\epsilon k T} (n_1 z_1^2 + n_2 z_2^2) = \frac{8\pi N e^2}{1000 \epsilon k T} I \quad \dots (4.12)$$

where I = ionic strength = $\frac{1}{2} \sum c_i z_i^2$ (c in mole/litre.)

$$\kappa = 50.29 \times 10^8 (\epsilon T)^{-1/2} \sqrt{I}$$

Equivalent Conductivities

$$\Lambda = \Lambda^0 - \left[\frac{2.801 \times 10^8 |z_1 z_2| q \Lambda^0}{(\epsilon T)^{3/2} (1 + \sqrt{q})} + \frac{41.25(|z_1| + |z_2|)}{\eta (\epsilon T)^{1/2}} \right] \sqrt{I} \quad \dots (7.29)$$

(Onsager limiting law for extreme dilutions).

$$\Lambda = \Lambda^0 - \left[\frac{2.801 \times 10^8 |z_1 z_2| q \Lambda^0}{(\epsilon T)^{3/2} (1 + \sqrt{q})} + \frac{41.25(|z_1| + |z_2|)}{\eta (\epsilon T)^{1/2}} \right] \frac{\sqrt{I}}{1 + \kappa a} \quad \dots (7.36)$$

(valid for moderate concentrations with suitable choice of a , especially for 1 : 1 electrolytes).

Formulae reduced for the case of 1 : 1 electrolytes:

$$\Lambda = \Lambda^0 - (B_1 \Lambda^0 + B_2) \sqrt{c} \text{ (Onsager limiting law.)}$$

$$\Lambda = \Lambda^0 - (B_1 \Lambda^0 + B_2) \sqrt{c} / (1 + B a \sqrt{c})$$

$$\text{or } \Lambda^0 = \Lambda + \frac{(B_1 \Lambda + B_2) \sqrt{c}}{1 + (B a - B_1) \sqrt{c}} \quad \dots (7.36)$$

(for moderate concentrations).

For values of B , B_1 and B_2 for aqueous solutions, see Appendix 7.1. In other solvents:

$$B = 50.29 (\epsilon T)^{-1/2} \times 10^8$$

$$B_1 = 8.204 \times 10^5 (\epsilon T)^{-3/2}$$

$$B_2 = 82.5 / [\eta (\epsilon T)^{1/2}] \text{ with } \eta \text{ in poise, } T \text{ in deg. K}$$

Transport Numbers (using first order electrophoretic terms only):

$$t_1 = \frac{\Lambda_1^0 - \frac{1}{2} |z_1| B_2 \sqrt{I} (1 + \kappa a)}{\Lambda^0 - \frac{1}{2} (|z_1| + |z_2|) B_2 \sqrt{I} (1 + \kappa a)} \quad \dots (7.39)$$

which reduces for 1 : 1 electrolytes to:

$$t_1 = \frac{\lambda_1^0 - \frac{1}{2}B_2\sqrt{c}/(1 + Ba\sqrt{c})}{\Lambda^0 - B_2\sqrt{c}/(1 + Ba\sqrt{c})} \dots (7.40)$$

with B, B_2 as given above.

Limiting law for transport numbers, valid at extreme dilution:

$$t_1 = t_1^0 + \frac{B_2}{2\Lambda^0} [(|z_1| + |z_2|)t_1^0 - |z_1|]\sqrt{I}$$

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APPENDIX 5.1

Specific Conductivities of Potassium Chloride Solutions (ref. 5.6)

Solution (demal)	g KCl/1000 g solution (in vacuo)	K_{sp} (int. ohm ⁻¹ cm ⁻¹)		
		0°	18°	25°
1 D	71.1352	0.06517 ₈	0.09783 ₈	0.11134 ₂
0.1 D	7.41913	0.007137 ₈	0.011166 ₇	0.012856 ₉
0.01 D	0.745263	0.0007736 ₄	0.0012205 ₂	0.0014087

APPENDIX 6.1

Limiting Equivalent Conductivities of Ions at 25° in water λ^0 in cm² Int. Ω^{-1} equiv.⁻¹

Ion	λ^0 *	Reference	Ion	λ^0	Reference
H ⁺	349.8 ₁	1, 2	OH ⁻	199.1 ₈	3
Li ⁺	38.6 ₉	2	F ⁻	55.4	4
Na ⁺	50.10	2, 5	Cl ⁻	76.35	5, 6
K ⁺	73.50	5, 6	Br ⁻	78.14	5, 6
Rb ⁺	77.8 ₁	7	I ⁻	76.8 ₄	6
Cs ⁺	77.2 ₄	7	N ₃ ⁻	69	8
Ag ⁺	61.9 ₀	2	NO ₃ ⁻	71.46	2
Tl ⁺	74.7	9	ClO ₃ ⁻	64.6	10
NH ₄ ⁺	73.5 ₅	11	BrO ₃ ⁻	55.7 ₄	12
CH ₃ NH ₃ ⁺	58.7 ₃	11a	IO ₃ ⁻	40.5 ₄	12c
(CH ₃) ₂ NH ₂ ⁺	51.8 ₇	11a	ClO ₄ ⁻	67.3 ₈	13
(CH ₃) ₃ NH ⁺	47.2 ₅	11a	IO ₄ ⁻	54.5 ₅	10
NMe ₄ ⁺	44.9 ₂	12	ReO ₄ ⁻	54.9 ₇	10
NEt ₄ ⁺	32.6 ₆	12	HCO ₃ ⁻	44.5 ₀	14
NPr ₄ ⁺	23.4 ₃	12	Formate	54.5 ₉	15
NBu ₄ ⁺	19.4 ₇	12	Acetate	40.9 ₀	16
NAm ₄ ⁺	17.4 ₇	12	Bromoacetate	39.2 ₂	16a
(CH ₃) ₃ (C ₆ H ₅)N ⁺	34.6 ₅	12a	Chloroacetate	42.2 ₀	16a
CH ₂ OH·CH ₂ ·NH ₃ ⁺	42.2 ₃	12b	Cyanoacetate	43.4 ₂	16b
Be ⁺⁺	45	4	Fluoroacetate	44.3 ₉	16a
Mg ⁺⁺	53.0 ₅	17	Iodoacetate	40.6 ₀	16b
Ca ⁺⁺	59.50	17, 19	Propionate	35.8	18
Sr ⁺⁺	59.4 ₅	17	Butyrate	32.6	18
Ba ⁺⁺	63.6 ₉	17	Benzoate	32.3 ₈	20
Cu ⁺⁺	53.6	21	Picrate	30.39	12
Zn ⁺⁺	52.8	21	SO ₄ ⁻	80.0 ₂	22
Co ⁺⁺	55	24	C ₂ O ₄ ⁻	74.1 ₅	23
Pb ⁺⁺	69.5	24a	CO ₃ ⁻	69.3	25
La ⁺⁺⁺	69.7	26	Fe(CN) ₆ ⁻⁻⁻	100.9	27
Ce ⁺⁺⁺	69.8	26	P ₃ O ₇ ⁻⁻⁻	83.6	28
Pr ⁺⁺⁺	69.6	26	Fe(CN) ₆ ⁻⁻⁻⁻	110.8	29
Nd ⁺⁺⁺	69.4	26	P ₄ O ₁₂ ⁻⁻⁻	93.7	28
Sm ⁺⁺⁺	68.5	26	P ₂ O ₇ ⁻⁻⁻	95.9	30
Eu ⁺⁺⁺	67.8	26	P ₃ O ₁₀ ⁵⁻	109	31
Gd ⁺⁺⁺	67.3	26	[Co(NH ₃) ₆] ⁺⁺⁺	101.9	27
Dy ⁺⁺⁺	65.6	26	[Co ₂ tri-en ₃] ⁶⁺	68.7	32
Ho ⁺⁺⁺	66.3	26	[Ni ₂ tri-en ₃] ⁴⁺	52.8	33
Er ⁺⁺⁺	65.9	26			
Tm ⁺⁺⁺	65.4	26			
Yb ⁺⁺⁺	65.6	26			

* The number of significant figures given is such that the last figure, if printed normally, is considered reliable within 1 or 2; and if printed below the line of the other figures, within about 5.

APPENDIX 6.1

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APPENDIX 6.2

Limiting Equivalent Conductivities, λ° , of Ions in Water at Various Temperatures

Ion	0°	5°	15°	18°	25°	35°	45°	55°	100°
H ⁺	225	250.1	300.6	315	349.8 ₁	397.0	441.4	483.1	630
OH ⁻	105	—	165.9 ₃	175.8 ₂	199.1 ₈	233.0 ₄	267.2 ₃	301.4 ₄	450
Li ⁺	19.4	22.7 ₆	30.2 ₀	32.8	38.6 ₈	48.0 ₀	58.0 ₄	68.7 ₄	115
Na ⁺	26.5	30.3 ₀	39.7 ₇	42.8	50.10	61.5 ₄	73.7 ₃	86.8 ₈	145
K ⁺	40.7	46.7 ₆	59.6 ₈	63.9	73.50	88.2 ₁	103.4 ₉	119.2 ₉	195
Rb ⁺	43.9	50.1 ₃	63.4 ₄	66.5	77.8 ₁	92.9 ₁	108.5 ₆	124.2 ₅	—
Cs ⁺	44	50.0 ₃	63.1 ₆	67	77.2 ₆	92.1 ₀	107.5 ₃	123.6 ₆	—
Ag ⁺	33.1	—	—	53.5	61.9 ₀	—	—	—	175
NH ₄ ⁺	40.2	—	—	63.9	73.5 ₆	88.7 ₃	—	—	180
N(CH ₃) ₄ ⁺	24.1	—	—	40.0	44.9 ₂	—	—	—	—
N(C ₂ H ₅) ₄ ⁺	16.4	—	—	28.2	32.6 ₈	—	—	—	—
N(C ₂ H ₅) ₃ ⁺	11.5	—	—	20.9	23.4 ₃	—	—	—	—
N(C ₂ H ₅) ₂ ⁺	9.6	—	—	—	19.4 ₇	—	—	—	—
N(C ₂ H ₅) ₁₁ ⁺	8.8	—	—	—	17.4 ₇	—	—	—	—
F ⁻	—	—	—	47.3	55.4	—	—	—	—
Cl ⁻	41.0	47.5 ₁	61.4 ₁	66.0	76.35	92.2 ₂	108.9 ₂	126.4 ₀	212
Br ⁻	42.6	49.2 ₅	63.1 ₆	68.0	78.1 ₄	94.0 ₃	110.6 ₈	127.8 ₆	—
I ⁻	41.4	48.5 ₇	62.1 ₇	66.5	76.8 ₄	92.3 ₉	108.6 ₄	125.4 ₄	—
NO ₃ ⁻	40.0	—	—	62.3	71.46	85.4 ₈	—	—	195
ClO ₄ ⁻	36.9	—	—	58.8	67.3 ₆	—	—	—	185
Acetate	20.1	—	—	35	40.9 ₀	—	—	—	—
Mg ⁺⁺	28.9	—	—	44.9	53.0 ₆	—	—	—	165
Ca ⁺⁺	31.2	—	46.9 ₈	50.7	59.50	73.2 ₄	88.2 ₁	—	180
Sr ⁺⁺	31	—	—	50.9	59.4 ₅	—	—	—	—
Ba ⁺⁺	34.0	—	—	54.6	63.6 ₉	—	—	—	195
Cd ⁺⁺	—	—	—	44.8	—	—	—	—	—
La ⁺⁺⁺	34.4	—	—	59.5	69.7 ₅	—	—	—	215
SO ₄ ⁻	41	—	—	68.4	80.0 ₂	—	—	—	260
Viscosity of water (centipoises)	1.787	1.516	1.138	1.053	0.8903	0.7194	0.5963	0.5044	0.2829

Note: The data at 25° are from references in Appendix 6.1. Those for OH⁻ from 15 to 55° are from the paper by Marsh, K. N. and Stokes, R. H., *Aust. J. Chem.*, 17 (1964) 740; $\lambda^\circ(\text{OH}^-) = 368.8_3$ at 75°; otherwise the data at 0° and 18° are from the limiting conductivities given in Landolt-Börnstein, based on $\lambda^\circ_{\text{Cl}^-}(0^\circ) = 41.0$, $\lambda^\circ_{\text{Cl}^-}(18^\circ) = 66.0$.

Data at 5°, 15°, 35°, 45° and 55° from papers of Gordon and his co-workers, and Owen *et al.* Those for the NH₄⁺ and NO₃⁻ ions at 35° are due to Campbell, A. N. and Bock, E., *Canad. J. Chem.* 36 (1958) 330.

Data at 100° from Landolt-Börnstein, readjusted to base $\lambda^\circ_{\text{Cl}^-}(100^\circ) = 212$.

The values at 100° are reliable only within several units; those at 5°, 15°, 25°, 35°, 45° to within the last figure given; those at 0° and 18° to within two or three units in the last figure given.

APPENDIX 6.3

Table 1

Equivalent Conductivity of Typical Electrolytes up to High Concentrations, in Aqueous Solution at 25°; c in mole/l; Λ in $\text{cm}^2 \text{int } \Omega^{-1} \text{equiv}^{-1}$

c	NaCl	KCl	BaCl ₂	LaCl ₃
0.0	126.45	149.85	139.98	146.0
0.0005	124.51	147.81	134.34	135.21
0.001	123.74	146.95	132.27	131.16
0.005	120.64	143.55	123.94	118.11
0.01	118.53	141.27	119.09	111.25
0.02	115.76	138.34	—	—
0.05	111.06	133.37	105.19	94.95
0.1	106.74	128.96	98.68	87.89
0.2	101.71	124.08	—	—
0.5	93.62	117.27	80.60	66.68
1	85.76	111.87	68.98	51.15
1.5	79.86	108.27	—	—
2	74.71	105.23	—	—
3	65.57	99.46	—	—
4	57.23	93.46	—	—
5	49.46	—	—	—
Ref.	1, 2	1, 2	3, 4, 7	5, 6

- ¹ SHEDLOVSKY, T., *J. Amer. chem. Soc.*, 54 (1932) 1411
² CHAMBERS, J. F., STOKES, J. M. and STOKES, R. H., *J. phys. Chem.*, 60 (1956) 985
³ JONES, G. and DOLE, M., *J. Amer. chem. Soc.*, 52 (1930) 2245
⁴ SHEDLOVSKY, T. and BROWN, A. S., *ibid.*, 56 (1934) 1066
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⁶ LONGSWORTH, L. G. and MACINNES, D. A., *ibid.*, 60 (1938) 3070
⁷ CALVERT, R., CORNELIUS, J. A., GRIFFITHS, V. S. and STOCK, D. I., *J. phys. Chem.*, 62 (1958) 47

APPENDIX 6.3

Table 2

Bibliography Recent Conductance Measurements in Concentrated Aqueous Solutions

Solute	Max. conc. mole/l.	Temp. °C	Ref.	Solute	Max. conc. mole/l.	Temp. °C	Ref.
HCl	9-12	5-65	1	KBr	3.75	0, 25	9*
LiClO ₄	19	25	2*	KI	6	25, 50	6
LiClO ₃	23.11†	131.8	2*	KH ₂ PO ₄	1.9	25	8
LiNO ₃	13.6	25	3*	NH ₄ Cl	5	25	10
LiNO ₃	14.4	110	3*	NH ₄ NO ₃	8	25	10
NaCl	5	25	5	NH ₄ NO ₃	11	25, 35	11*
NaCl	5	50	6	NH ₄ NO ₃	15	95	16*
NaI	10	0, 30, 50	7*	NH ₄ NO ₃	18.0†	180	12*
NaClO ₃	10	0, 30, 50	7*	AgNO ₃	8	25, 35	11*
NaCNS	10	0, 30, 50	7*	AgNO ₃	14	95	16*
Na ₂ HPO ₄	3.9	25	8	AgNO ₃	23.19†	221.7	12*
KCl	4	25	5	H ₂ SO ₄	18	50, 75	13*
LiNO ₃ in EtOH and EtOH-H ₂ O	3-11	25	4*	H ₂ SO ₄	18	25-155	14
				H ₃ PO ₄	18	25	8*
				K ₂ Fe(CN) ₆	1	25	15
				K ₄ Fe(CN) ₆	0.7	25	15
				MgSO ₄	2.9	25	15
				HCOOK	6.5	50.5	17*
				HCOOCs	10	50.5	17*

* These papers include viscosity data.

† Fused salt.

For work at lower concentrations, see references to Table 1; for extensive earlier work, usually of lower accuracy, see *Int. crit. Tab.*, Vol. VI, pp. 230-256.

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¹² CAMPBELL, A. N., KARTZMARK, E. M., BEDNAS, M. E. and HERRON, J. T., *ibid.*, 32 (1954) 1051
¹³ CAMPBELL, A. N., KARTZMARK, E. M., BISSET, D. and BEDNAS, M. E., *ibid.*, 31 (1953) 303
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¹⁶ CAMPBELL, A. N., KARTZMARK, E. M., *Can. J. Chem.*, 30 (1952) 128
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of linear superposition. For since outside the Bjerrum region $z_j e \psi_j < 2kT$, n_j^+ cannot by equation (4.18) fall below zero. Thus once more we reach the conclusion that the theory is only really adequate for symmetrical electrolytes. The use of the modified distribution function proposed by Eigen and Wicke offers no solution of this particular difficulty, for as Figure 4.2 shows, the greatest values of the vital quantity $\frac{z_j e \psi_j}{kT}$ are attained in dilute solutions,

where the extra factors introduced by Eigen and Wicke make no material difference. We may, however, conclude from a study of Figure 4.2 that, since the Debye-Hückel treatment of the potential problem is known from experiment to be successful for dilute solutions of fully dissociated symmetrical electrolytes, it should, if anything, be more so in more concentrated solutions; and for 1:1 electrolytes, at least, it does not lead to any physically absurd distributions such as negative concentrations of ions, or concentrations too high to be consistent with the known sizes of the ions. We therefore intend to use the Debye-Hückel expression (4.13) for the potential as a basis for all our theoretical calculations, along with the distribution function (4.14) for symmetrical electrolytes and with the less adequate (4.9) for unsymmetrical ones.

The alternative distribution functions which we have discussed offer no improvement in respect of the self-consistency of the resulting theory, but have the disadvantage of adding considerably to the complexity of the formulae.

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THE MEASUREMENT OF CONDUCTIVITIES AND TRANSPORT NUMBERS

METHODS OF MEASURING ELECTROLYTIC CONDUCTIVITY

THE requirements for precise measurement of electrolytic conductivity may be summed up as (a) accurate temperature control, (b) avoidance of polarization at the electrodes, and (c) accuracy in the electrical measurements themselves.

As regards temperature control, the majority of aqueous electrolytic solutions have a temperature coefficient of conductivity close to 2 per cent per degree at 25°. That of the hydrogen ion is appreciably lower, about 1.4 per cent per degree at 25°. If an accuracy of 0.01 per cent is sought, the thermostat should therefore be capable of controlling the temperature to $\pm 0.005^\circ$ or better throughout the measurements. When, as is frequently the case, the solution studied has a temperature coefficient similar to that of the standard potassium chloride solution used for the cell calibration, it is less important to know the exact temperature with the same accuracy provided that it is constant, for a constant error of a few hundredths of a degree will largely be compensated by a corresponding change in the conductivity of the standard. This, of course, does not apply when the temperature coefficients differ appreciably or when the measurements and the cell calibration are made at different temperatures; in such cases sensitive and recently-calibrated thermometers, or preferably a platinum resistance thermometer, are desirable. The use of water as a thermostat liquid should be avoided owing to undesirable capacity effects across the cell walls in a.c. measurements and to the risk of electrical leakage currents in d.c. measurements. A light paraffin such as kerosene is a satisfactory thermostat medium at ordinary temperatures. The errors caused by the use of water as a thermostat liquid were thoroughly investigated by JONES and JOSEPHS⁽¹⁾, who found differences of up to 0.5 per cent between the resistance in an oil- and a water-filled thermostat. The errors varied in a complicated manner with the cell design, the conductivity of the thermostat water, and the resistance of the cell being measured; they were greater at higher frequencies

and with high cell resistances, indicating that they arose mainly from capacitance by-paths through the cell walls and the thermostat water. Earthing the thermostat tank changed the sign of the errors, and reduced their magnitude, but by no means eliminated them. Rather surprisingly, they were smaller when the conductivity of the thermostat water was increased by the addition of potassium chloride.

Polarization errors are usually minimized by the use of audio frequency alternating current for the measurements and by coating the electrodes with a heavy deposit of platinum black, a system initiated by KOHLRAUSCH⁽²⁾. While this procedure correctly applied is undoubtedly effective, the use of alternating current enormously complicates the electrical technique required for high-accuracy measurements, owing to the need for compensation of capacitive and inductive effects in the circuit. The only alternative is, however, the use of electrodes which are truly reversible to one of the ions in the solution; this permits the use of the simpler direct current measuring techniques. Though the d.c. method has received increasing attention in recent years the conventional a.c. method seems likely to remain the standard one for general applications.

ALTERNATING CURRENT CONDUCTANCE MEASUREMENTS

In the simple Wheatstone bridge (Figure 5.1) used for d.c. resistance measurements, the galvanometer shows no deflection at balance; the potentials at *A* and *B* are therefore equal, whence $R_1/R_2 = R_3/R_4$. In the a.c. bridge (Figure 5.2) the battery is replaced by a sinusoidal alternating potential from an oscillator, and the galvanometer by a suitable detector. The condition for balance (*i.e.*, no signal in the detector) is that the alternating potentials at *A* and *B* are of equal amplitude and exactly in phase, which leads to the relation $Z_1/Z_2 = Z_3/Z_4$, where the impedance Z is the a.c. analogue of resistance.

Impedance is conveniently represented as a complex quantity having the following properties: (i) impedances combine like resistances, *i.e.*, impedances in series add, while if in parallel their reciprocals (admittances) add; (ii) a pure resistance R has an impedance $Z = R$ which is entirely a real quantity; (iii) a perfect condenser of capacity C has impedance $Z = 1/(j\omega C)$ where ω is angular frequency and j is an operator having the mathematical properties of $\sqrt{-1}$, which represents a phase displacement of 90° between current and potential; (iv) a pure inductance L has impedance $Z = j\omega L$. This representation of impedances by complex numbers

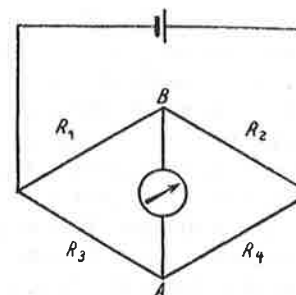


Figure 5.1

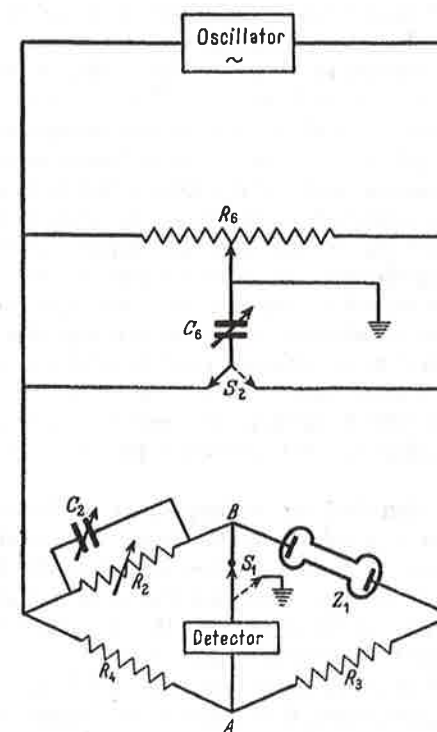


Figure 5.2. Basic circuit of a.c. bridge for measurement of electrolytic conductance

has considerable utility since equality of two impedances demands equality of both real and imaginary parts, so that amplitudes and phases in any part of an a.c. network can in principle be calculated by methods formally similar to those used in d.c. work.

The design and construction of high-precision conductance bridges was studied intensively by GRINNELL JONES⁽³⁻⁹⁾ and his colleagues and also by SHEDLOVSKY⁽¹⁰⁾ and the principles they laid down are the basis of today's designs. Referring to *Figure 5.2*, the ratio-arms R_3 and R_4 are made equal (usually 1000 ohms) and of identical construction so that the residual capacities between turns and between the coils and nearby objects are exactly equal. The usual device of winding with doubled wire ensures that the inductances of all resistance coils in the bridge are negligible at audio-frequencies. The measuring-arm R_2 may be a separate resistance-box but is usually built into the same box as R_3 and R_4 . In parallel with R_2 is a variable capacitor C_2 of maximum capacity 0.001 microfarad; this is necessary to obtain a sharp balance-point since the cell will in general have an impedance Z_1 which is not purely resistive. A further valuable aid to a sharp balance-point is the 'Wagner earth' comprising R_6 and C_6 ; the object of these components, which are used in conjunction with the earthing-switch S_1 , is to ensure that at balance the potentials at *A* and *B* are not merely equal but are actually earth-potential so that pick-up of hum and stray noise by the detector is minimized. A rough balance is first obtained with S_1 in the position shown by the full line; then S_1 is turned to earth and R_6 and C_6 are adjusted to give a minimum signal in the detector; S_1 is then restored to its original position and a final balance is obtained. IVES, PRYOR and FEATES⁽¹¹⁾ obtain an equivalent effect by the use of two 1000 Ω radio potentiometers connected across the output of the oscillator, the moveable tappings being taken to the bridge input.

The oscillator and detector are important auxiliaries. The oscillator should give a good sinusoidal wave-form at all frequencies from 500 c/s to several thousand. The amplitude should be variable from a few volts down to very low values and both the output terminals should be isolated from earth. It is best to isolate the bridge from both the oscillator and the detector by good quality transformers as otherwise the Wagner earth will not function properly. The detector consists first of one or two stages of amplification, in which an automatic gain control may be incorporated to limit the maximum signal when the bridge is far off balance. The amplifier may be followed by a telephone headset, which is remarkably sensitive around 1000 c/s, but the tendency today is to use a cathode

ray oscilloscope which is much more versatile and less nerve-racking. The horizontal deflection plates are fed from the oscillator and the vertical ones from the amplifier; the out-of-balance trace is an ellipse which becomes a horizontal straight line at balance. An outfit of this kind can readily detect a variation of a few parts in a million in the bridge setting.

Cell design. The object of the measurements is to determine the pure ohmic resistance R_1 of the solution between the electrodes. If the cell impedance Z_1 consisted only of this resistance, R_1 would equal R_2 at all frequencies and the capacitor C_2 would be required only to compensate for capacity between the cell leads and for the small capacity in parallel with the cell due to the action of its electrodes as a capacitor with the solution as dielectric. In practice there are several other sources of impedance which cause R_2 and C_2 to vary appreciably with frequency. Some of these can be avoided by proper design and the others are inherent in the electrode processes. The former comprise the Parker effect and the effect of a conducting thermostat medium discussed on p. 87. The Parker effect arises when the cell leads pass near to the cell solution giving the effect of a capacitor connected between one end of R_1 and some point in the middle of R_1 ; it can be avoided by spacing the leads well away from parts of the cell containing solution as in the designs shown in *Figure 5.3*. Mercury-filled lead-in tubes are often used but are a nuisance: we replace them by heavy silver wires welded to the outer ends of the electrodes beyond the seals.

Designs for electrode assemblies which can be used as dipping electrodes in containers of any size are given by BRODY and FUOSS⁽¹²⁾.

The effects associated with the electrode processes themselves are of interest and their understanding is necessary in order to eliminate them from the measurements. KOHLRAUSCH⁽²⁾ showed that this could be largely achieved by coating the electrodes with platinum-black, when R_2 becomes practically independent of frequency; this course is, however, not always possible since platinum-black may catalyse unwanted reactions and may in dilute solutions adsorb appreciable quantities of solute, making necessary emptying and refilling of the cell until a constant reading is obtained. The platinizing solution recommended by JONES and BOLLINGER⁽⁸⁾ is 0.025N hydrochloric acid containing 0.3 per cent of platinic chloride and 0.025 per cent of lead acetate; the lead acetate improves the adherence of the deposit. The platinizing current should be 10 mA/cm², the polarity being reversed every ten seconds. Even a barely visible deposit greatly reduces the frequency dependence and a deposit corresponding to a few coulombs/cm² is ample.

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Electrode effects can also be eliminated by eliminating the electrodes. This is done physically in the transformer bridge due to CALVERT⁽¹³⁾ *et al.*, in which a loop of solution links a current transformer with a voltage transformer; and virtually by the double cell

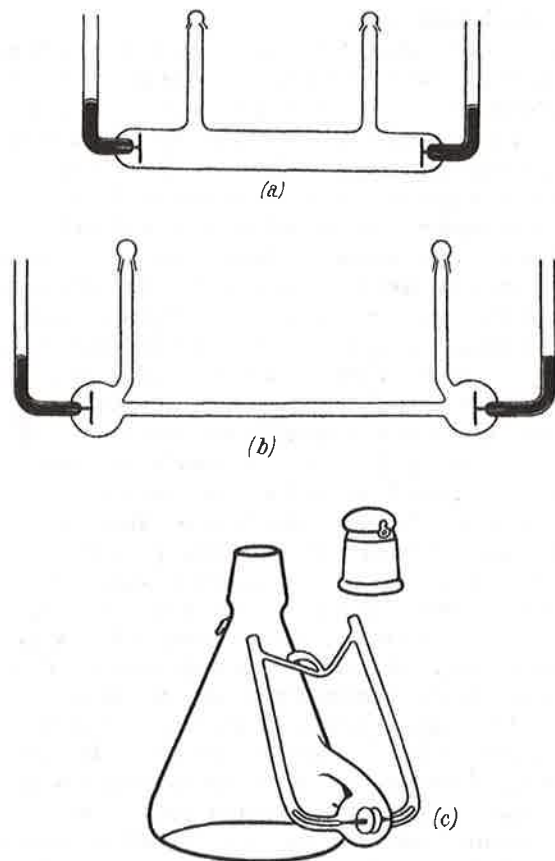


Figure 5.3. Typical conductivity cell designs for (a) moderate, (b) high, and (c) low concentrations; the last after DAGGETT, H. M., BAIR, E. J. and KRAUS, C. A., *J. Amer. chem. Soc.*, 73 (1951) 799

design of FEATES, IVES and PRYOR⁽¹¹⁾ who use two cells with identical electrodes but with different lengths of solution between them and measure the difference in the resistances of the two cells, this difference showing only a small residual frequency dependence. They also employ two independent leads to each electrode so that lead

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resistances can be completely eliminated by the 'four leads' method used in platinum resistance thermometry. By these means they have attained remarkable accuracy in the conductometric measurement of ionization constants.⁽¹⁴⁾

The elimination of electrode effects from ordinary cells, without platinization, is still of great importance. The modern theory of electrode processes leads to a schematic representation of the conductance cell shown in Figure 5.4, which is that proposed by IVES' school⁽¹¹⁾ with the addition of the 'Warburg impedance' -W-. In this

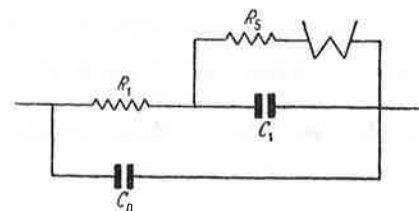


Figure 5.4. Network electrically equivalent to conductance cell

figure R_1 is the true ohmic resistance of the electrolyte, which is to be determined. This is independent of frequency at audio frequencies, because the Falkenhagen effect, associated with the relaxation of the ionic atmospheres, does not become appreciable until radio frequencies are attained. In series with R_1 is the capacity C_1 of the double layer of ions at the electrode surfaces: this is also expected to be independent of frequency. Because of the small thickness of the double layer, this capacity is surprisingly large, often amounting to several microfarads per sq. cm. of electrode surface. The current through R_1 is transported across the double layer mainly by virtue of this capacity without any actual discharge or formation of ions, for the cell gives definite resistance readings when the potential across it is only a few millivolts, far too little to cause electrolysis of most solutions at bright platinum electrodes. However, some electrolysis will normally occur simultaneously, perhaps through the depolarizing action of dissolved oxygen and the discharge of ions of the solvent, and in some cases through reversible discharge of ions of the electrolyte, for example in a cell with silver electrodes in a solution of silver nitrate. The electrolysis process is represented as a 'faradaic leakage' in parallel with the double layer. In general, as shown by GRAHAME⁽¹⁵⁾ and by RANGLES,⁽¹⁶⁾ it will consist of two parts: a pure resistance R_2 , independent of frequency, and a 'Warburg impedance' at the electrodes. For the full theory

of the Warburg impedance, the original literature should be consulted: here we merely note that it can be regarded as equivalent to a resistance and a capacity in series, the impedance of both being the same at any one frequency but both varying inversely as $\omega^{\frac{1}{2}}$. They can therefore be represented together by:

$$-W- = k(1 - j)/\sqrt{\omega}$$

where k is a constant of dimensions (resistance $\times$ time $^{-\frac{1}{2}}$). Solution of the balance conditions leads to the result that, if the impedance of the cell arm with the omission of C_0 is denoted by Z , then:

$$1/R_2 = \text{real part of } Z^{-1}$$

from which R_2 may be determined in terms of R_1 , C_1 , R_5 and k . The following special cases are of interest.

(i) R_5 infinite: the electrodes are ideally polarized and:

$$R_2 = R_1 + (\omega^2 C_1^2 R_1)^{-1} \quad \dots (5.1)$$

This is unlikely to arise in practice, as pointed out by Ives, Pryor and Feates, because of the depolarizing action of dissolved oxygen: it might perhaps be expected in solvents of very low self-dissociation and with an electrolyte whose ions have high discharge potentials.

(ii) If the Warburg impedance is negligible compared to R_5 ,

$$R_2 = R_1 + \frac{R_1 R_5 + R_5^2}{R_1(1 + \omega^2 C_1^2 R_5^2) + R_5}$$

which, since in any ordinary conductance cell $R_1 \gg R_5$, approximates well to:

$$R_2 = R_1 + R_5/(1 + \omega^2 C_1^2 R_5^2) \quad \dots (5.2)$$

This is the model proposed by Feates, Ives and Pryor as applicable to a conductance cell with grey platinized electrodes; it also corresponds to the behaviour of a cell with bright platinum electrodes in an aqueous solution. Although equation (5.2) is ill-adapted to graphical extrapolation to infinite frequency, it is found in practice⁽¹⁷⁾ that R_1 as obtained by solving it for three frequencies agrees well with the value obtained by a linear extrapolation of R_2 against ω^{-1} .

(iii) If C_1 is very large, so that its impedance is small compared with that of the faradaic leakage, $R_2 = R_1$ at all frequencies. A very near approach to this behaviour is found with heavily blacked platinum electrodes.

(iv) Where the Warburg impedance is large compared to R_5 but small compared to R_1 , one obtains solutions approximating to:

$$R_2 = R_1 + k/\sqrt{\omega} \quad \dots (5.3)$$

This corresponds to the behaviour found by JONES and CHRISTIAN⁽⁷⁾ in their studies of electrode polarization, more particularly at silver electrodes in silver nitrate solution or at platinum electrodes in acid solutions. It does not appear to be as general as was thought earlier for bright platinum electrodes: Jones and Christian themselves found marked curvature of the R_2 vs. $1/\sqrt{\omega}$ curves for such electrodes in potassium chloride solution and BRODY and FUOSS⁽¹²⁾ report a number of cases in which R_2 is quadratic in $k/\sqrt{\omega}$. These are presumably intermediate between (ii) and (iv).

To sum up, it is necessary when using bright platinum electrodes to measure at a number of frequencies, preferably including higher frequencies than the usual 2 Kc/s., and to extrapolate to infinite frequency according to the kind of frequency dependence observed.

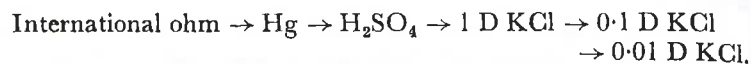
Standards of specific conductance—The actual measurements are of the resistance between two electrodes of fixed shape and size in a cell filled with the solution. This resistance naturally depends on the geometry of the cell as well as on the dimensions and separation of the electrodes; it is therefore the invariable practice to calibrate the cell by means of a solution of known specific resistance. It is usual to define a cell constant a by

$$K_{sp} = a/R,$$

R being the measured resistance with a solution of specific conductivity, K_{sp} , in the cell. To provide such a standard, a great deal of careful work has been expended on determinations of the specific resistance of potassium chloride solutions. The standards generally accepted today are those of JONES and BRADSHAW⁽⁶⁾, and in view of their fundamental importance the methods used in obtaining them will now be described.

At the time the work was done, the accepted unit of electrical resistance was the international ohm, defined as the d.c. resistance at the ice-point of a uniform column of mercury 106.300 cm in length and of 14.4521 g mass. Thus a conductivity cell could be calibrated in international ohm units by measuring its resistance when filled with mercury at 0°C. Jones and Bradshaw prepared cells having resistances of approximately one ohm when filled with mercury at 0°C, and measured their resistance (to direct current) on a Kelvin bridge in terms of the international ohm. However, it was not possible to use these cells directly for determining the specific conductivity of standard potassium chloride solutions, since a one molar solution at 0° would have had a resistance of the order

of 100,000 Ω , which is too high for accurate determination on an a.c. bridge. They therefore measured the conductivity at 0° of a relatively concentrated (6 N) sulphuric acid solution, which was then used to calibrate smaller cells. In these smaller cells, the specific conductivity of one demal solutions (see below) was measured at 0°, 18° and 25°, with allowance for the thermal expansion of the cell at the higher temperatures. The one demal potassium chloride solution was then used to calibrate still smaller cells in which the conductivity of 0.1 demal potassium chloride was measured and finally the 0.1 demal solution was used to calibrate even smaller cells in which the conductivity of 0.01 demal potassium chloride was found. Thus the series of steps by which the standard values for potassium chloride were obtained may be represented:



In these measurements, the exact concentration of the sulphuric acid was not required as it served only as an intermediate standard; but the concentration of the potassium chloride was very carefully defined in terms of the weight of salt in 1,000 g of solution. Appendix 5.1 gives the compositions and specific conductivities of the three standard solutions determined by Jones and Bradshaw.

The term 'demal' was introduced by PARKER and PARKER⁽¹⁸⁾ in an earlier determination of standards; it is not in general use as a measure of concentration, but the name was retained by Jones and Bradshaw as a convenient label for their standard compositions. It will be noted that Jones and Bradshaw's standards, being defined only in terms of weights *in vacuo*, are independent of volume standards and of atomic weights, changes in both of which have caused considerable confusion ever since the first standards were proposed by Kohlrausch. It is particularly unfortunate that the extensive compilation of electrolytic conductivities in the International Critical Tables is in terms of the earlier Parker and Parker standards, which are now generally considered to be unsatisfactory. Practically all recent work, in English-speaking countries at least, has, however, been based on the Jones and Bradshaw standards (Appendix 5.1), and for the sake of consistency they should be retained even if future work shows them to be slightly in error. Already one change has occurred which emphasizes the difficulty of defining a standard to a high degree of accuracy: the international ohm is no longer the recommended unit of resistance, having been replaced by the absolute ohm which is defined in terms of the fundamental units of

the c.g.s. electromagnetic system. The relation between the absolute and international ohm is:

$$1 \text{ int. ohm} = 1.00050 \text{ abs. ohm.}$$

It follows that the measure of a given resistance in absolute ohms is 0.050 per cent larger than its measure in international ohms; and the measure of a given specific conductivity in $(\text{abs. ohm})^{-1} \text{ cm}^{-1}$ is 0.050 per cent smaller than in $(\text{int. ohm})^{-1} \text{ cm}^{-1}$.

However, there seems to be no point in revising all the literature values of conductivities of electrolytes into $(\text{abs. ohm})^{-1} \text{ cm}^{-1}$, as we are far more concerned with the *variation* of conductivity with concentration than with its exact value to five significant figures; theory is unable to predict conductivities *a priori* to even two significant figures, though it can do much better with the *change* of conductivity with concentration. The fact that resistance bridges are now calibrated in absolute ohms need not cause any difficulty, as the usual experimental determination is not that of an actual specific conductance, but its ratio to the specific conductance of one of the standard solutions, *via* the cell constant. Hence as long as the standard is defined in international units, the conductivity of the substance studied will be in the same units. It must be noted that the standard specific conductivities recorded in Appendix 5.1 are corrected for the specific conductivity of the water used in preparing them. As this is usually of the order of $1 \times 10^{-6} \Omega^{-1} \text{ cm}^{-1}$, it will not be significant for the 1 demal standard, but must be allowed for when the 0.1 D and especially the 0.01 D solutions are used for cell calibration; in the last it can make a difference of the order of 0.1 per cent to the cell constant.

Variation of the Cell-constant with Temperature

The standard solutions specified by Jones and Bradshaw have accurately known conductivities at 0°, 18°, and 25°. For work at other temperatures, the cell constant measured at one of the standard temperatures must be adjusted slightly to allow for expansion of the glass and the platinum electrodes. The naïve expectation that the correction factor would be the same regardless of the geometry of the cell is disappointed, as the following argument shows.

Treating the cell as consisting of a number of regions in each of which the current density is uniform, the cell constant is given by

$$a = \sum \frac{l}{A}$$

where l is the length of each region and A its cross-sectional area

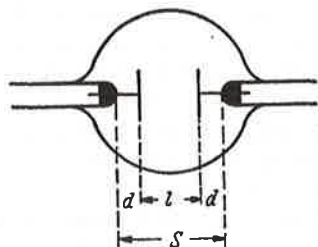
5 MEASUREMENT OF CONDUCTIVITIES AND TRANSPORT NUMBERS

normal to the current. Two extreme types of cell design may now be considered:

- (i) A long narrow tubular cell with large electrodes in bulbs at the ends. Here nearly all the resistance is contributed by the narrow tube. If the glass has a linear expansion coefficient α_g , the relative change in a with temperature (t) is given by:

$$\begin{aligned}\frac{1}{a} \frac{da}{dt} &\approx \frac{1}{l} \frac{dl}{dt} - \frac{1}{A} \frac{dA}{dt} \\ &= \alpha_g - 2\alpha_g = -\alpha_g\end{aligned}$$

- (ii) A cell consisting of two large electrodes of area A separated by a small distance l . The electrodes are supported by platinum wires sealed into the cell at points distant S apart,



the length of wire between the electrode and the seal being d in each case so that the separation of the electrodes is $l = S - 2d$. Thermal expansion has three distinct effects:

- The area A of the electrodes is increased.
- The distance S is increased.
- The distances d are increased.

The cell constant is given approximately by:

$$a = \frac{l}{A} = \frac{S - 2d}{A}$$

By logarithmic differentiation with respect to temperature one obtains the temperature coefficient:

$$\frac{1}{a} \frac{da}{dt} = \alpha_g \frac{S}{S - 2d} - 2\alpha_{Pt} \frac{S - d}{S - 2d}$$

ALTERNATING CURRENT CONDUCTANCE MEASUREMENTS

If the expansion coefficients of the glass and the platinum are equal, as is approximately the case with a soda-glass cell, we have

$$\frac{1}{a} \frac{da}{dt} = -\alpha_g$$

just as for the long tubular cell. In this case therefore the expansion correction is the same for both, and the result clearly generalizes to cells of any shape. But if Pyrex glass is used, differences can arise. Taking

$$\alpha_g = 3.6 \times 10^{-6} \text{ deg C}^{-1}, \text{ and } \alpha_{Pt} = 9 \times 10^{-6} \text{ deg C}^{-1}$$

we have for the two cases:

$$(i) \frac{1}{a} \frac{da}{dt} = -3.6 \times 10^{-6} \text{ deg C}^{-1}$$

$$(ii) \text{ Putting } S = 10 \text{ mm and } d = 2 \text{ mm}$$

$$\frac{1}{a} \frac{da}{dt} = -18 \times 10^{-6} \text{ deg C}^{-1}$$

so that for a temperature change of 100° the cell constants would change by 0.04% and 0.18% respectively.

A very large temperature coefficient may occur if the electrodes are very close together and supported by rather long wires sealed into Pyrex glass: thus for example if $S = 20$ mm and $d = 9$ mm, giving electrodes only 2 mm apart supported on 9 mm lengths of wire:

$$\frac{1}{a} \frac{da}{dt} = 63 \times 10^{-6} \text{ deg C}^{-1}$$

corresponding to a change of 0.63 per cent in the cell constant for 100° change.

Direct Current Conductivity Measurements

It will be clear from the foregoing account that the alternating current method for conductivity measurements, though capable of extreme accuracy, introduces a great many new complications due to capacity effects in the circuit; the compensating advantages are the elimination of polarization, the fact that electronic amplifiers are easily incorporated in the detector circuit and that thermoelectric effects and contact potentials in the resistance box are unimportant.

The direct current method is therefore simpler in principle, requiring only the passage of a steady current through the solution

and through a standard resistance in series, and the comparison of the potential developed between two fixed points in the solution with that across the standard resistance. Since potentiometric measurements can be made with an accuracy of 0.001 per cent the method should be capable of accuracy comparable with that of the best a.c. technique. It is, however, essential that the potential measurements in the solution be made between strictly reversible

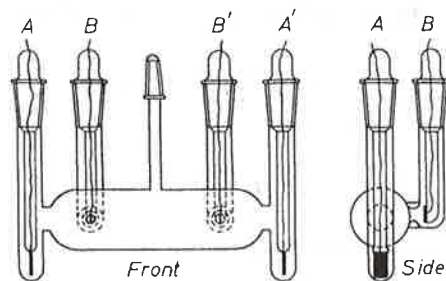


Figure 5.5. GORDON's d.c. conductivity cell

electrodes. The greatest success with the d.c. method to date has been achieved by GORDON and his collaborators⁽¹⁹⁾ on dilute halide solutions in water and in methanol.

A slightly modified form of their cell, described by ELIAS and SCHIFF^(19a), is shown in Figure 5.6.

A cylindrical Pyrex tube, about 20 cm in length and 5 cm in diameter, has two side-tubes about 10 cm apart to hold the probe-electrodes, *B*, *B'*, each of which is made of an 8 mm platinum disc, so mounted that its position is not subject to appreciable variation in successive experiments. Each electrode is covered with fused glass except for a narrow slip 1×6 mm in size which is silver-plated and chloridized (or bromidized). The electrodes, *A*, *A'*, which introduce the current are inserted in narrow collimating tubes at each end: they are of heavy silver-plated platinum, dipped in fused silver chloride (or bromide). Through this cell, with a calibrated 500 Ω resistance in series, current is passed by means of the constant current circuit used in transport number work. The potential across the 500 Ω resistance is measured first, then the potential across the probe-electrodes; to eliminate any bias, it is advisable to reverse the current and measure the potential again, with a further measurement across the standard resistance as a check.

In Gordon's design the fact that there is a potential gradient near the potential-measuring electrodes makes it essential that these be small and reproducibly located, so that only silver-silver halide

electrodes are really suitable. IVES and SWAROOPA^(19b) used a cell in which the solution being measured was connected to two quinhydrone electrodes via arms so placed that no potential gradient occurred in them; this made possible the use of a liquid junction between the quinhydrone electrode and the solution, increasing the scope of the method. ELIAS and SCHIFF^(19a) have developed a more precise apparatus on the same principle. Their cell is that of Figure 5.5, but the tubes *B* and *B'* carrying the potential-measuring electrodes are replaced by the components shown in Figure 5.6.

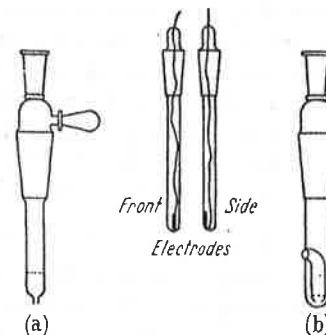


Figure 5.6. Liquid junction electrode vessels. After ELIAS, L. and SCHIFF, H. I., *J. phys. Chem.*, 60 (1956) 595

Silver-silver halide electrodes mounted in the small central tubes are surrounded by alkali halide solution in tubes *a* or *b*, this solution making a liquid junction with the main cell solution at the position shown by the dotted lines. Type *a* is used when the cell solution is denser than the probe solution and type *b* in the opposite case. Since no electrolysis current flows through the solution in the side arm, the potential between the probe electrodes is proportional to the resistance of the cell solution. The current carrying electrodes are silver plated platinum; no electrode is needed reversible to either ion of the electrolyte being studied and the method has general application.

Radio Frequency Measurement of Conductivity

The audio frequency a.c. bridge technique and the d.c. methods discussed above are high precision methods designed to meet the exacting requirements of the physical chemist whose concern is with the nature of electrolyte solutions and the mathematical interpretation of their behaviour. Conductivity measurement is, however, also an everyday analytical tool of immense value; for purposes

such as conductimetric analysis it is seldom necessary to adopt all the refinements of technique described above. Many commercial instruments for conductimetric titration, for example, operate at the rather low frequency (~ 50 c/sec) of the electric mains, with an accuracy of one or two per cent, which is quite adequate for the purpose. A very interesting development of recent years, with great possibilities for analytical and process-control applications, is the use of radio-frequency methods for conductivity measurements. The great virtue of these methods is that the electrodes need not be in contact with the solution; polarization errors are therefore completely absent. The cell vessel may be a simple test-tube or flask which is placed either within a coil or between the plates of a condenser which form the elements of an oscillatory circuit. The presence of the electrolytic resistance alters the frequency of the oscillations, or in another method, the coupling between two oscillatory circuits, and the change is measured by suitable meters.

MEASUREMENT OF TRANSPORT NUMBERS

The experimental methods available for measuring transport numbers fall into three categories: (i) the Hittorf method, (ii) the moving boundary method, and (iii) a method depending on concentration cells with a liquid junction.

The first of these is so familiar that it requires little comment. Devised in 1853, it was the instrument for an outstandingly comprehensive study lasting over half a century and, although it has been superseded by other techniques, the value of this one man's contribution should be recognized. It is especially remarkable that many of these measurements were made before the Arrhenius ionic theory was developed. Several modifications of the apparatus have been made, but all consist essentially of an anode compartment, a cathode compartment and a third intervening compartment. Current, in amount measured by a coulometer, is passed and the change in composition of each section determined analytically; assuming that the current is not passed so long that the composition of the central compartment changes, then the loss in either the anode or the cathode compartment gives one of the two transport numbers.

The application of the Hittorf method is limited by two main factors: at least one and preferably both the electrodes must be reversible and extreme accuracy is needed in the analysis of the solution before and after electrolysis. There are few electrodes through which it is possible to pass a substantial quantity of electricity, e.g., 20 coulombs, without gassing or other unwanted side-reactions. Well annealed, very pure silver behaves satisfactorily in

aqueous silver nitrate solution provided oxygen is excluded; silver-silver halide electrodes are suitable for aqueous chloride and bromide solutions. The possibility of electrode reactions with the solvent must be considered. The analytical problem can be reduced in importance by passing sufficient electricity to produce a large change in concentration, but this involves either long runs with the danger of diffusion of the anode and cathode solutions into one another, or large currents with consequent overheating of the solution, resulting in turbulent mixing.

Figure 5.7 shows a modern form of Hittorf apparatus developed by STEEL and STOKES⁽²⁰⁾ for use with alkali bromide solutions in

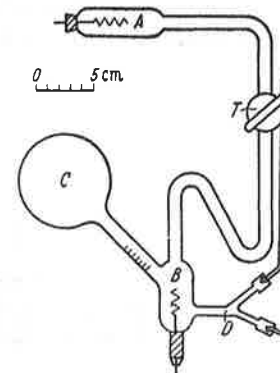


Figure 5.7. Diagram of transport number apparatus after STEEL, B. J. and STOKES, R. H., *J. phys. Chem.*, 62 (1958) 450

mixed solvents. The important feature of this apparatus is the built-in conductance-cell which makes possible the analysis of the cathode solution without removal from the apparatus. After an initial measurement of the conductance, current is passed through the silver-silver bromide electrodes to produce a 10 to 20 per cent change in concentration of the solution near the electrodes. The tap is then closed, the whole of the cathode solution is thoroughly mixed in the bulb and returned to the position shown for a second conductance measurement. As a check on the reversibility of the electrode-reactions, the tap is then opened, the anode and cathode solutions are remixed in the bulb, and a final conductance reading—which should agree with the first—is taken. The volume of the cathode solution is read off from the calibrated stem of the mixing-bulb; to ensure proper drainage, the bulb is coated with a water-repellent silicone film. The 'apparent' transport number of the cation is $v\Delta cF/Q$, where v is the volume of the cathode solution, Δc is the change in its concentration in equivalents per unit volume and Q

is the quantity of electricity passed; this is converted to the Hittorf frame of reference, (motion relative to the solvent), by the use of density data. Some results obtained by this method will be considered in Chapter 11; while not quite equal in precision to the moving-boundary method, it offers some prospect of success in non-aqueous solutions where transport number data are urgently needed.

The Moving Boundary Method

The considerations upon which this method is based are simple: let solutions of two salts (having a common anion, X^-), form a boundary ab and let current be passed so that the cations move up the tube and the anions down the tube (Figure 5.8). If the conditions are chosen properly the boundary will remain distinct but will move

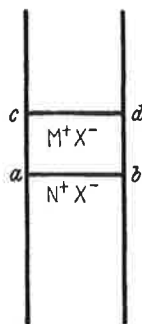


Figure 5.8

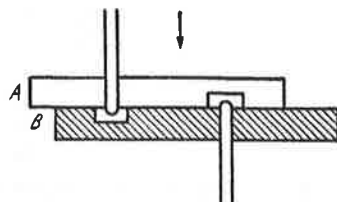


Figure 5.9

up the tube. After a certain time let it be at the position cd . In this interval all the cations, M^+ , in the volume V between cd and ab must have crossed a plane at cd . If the amount of electricity passing be Q coulombs, then the amount moving upwards is $t_1 Q$ coulombs. If V is the volume between ab and cd and the concentration of the solution of M^+X^- is c ion equivalents of M^+ per unit volume, the amount of electricity moving upwards must be $V[c]F$ whence:

$$t_1 = VcF/Q$$

All variants of this method depend fundamentally on measuring this volume for unit amount of electricity passed and the successful application of the method depends on three factors: (1) the construction of an apparatus capable of producing a sharp boundary, (2) the use of a suitable salt N^+X^- (called the indicator) and its use at the proper concentration, and (3) a small correction for changes in the position of the boundary due to volume changes. The sharp boundary can be produced by one of three methods. The first was

developed by MACINNES and BRIGHTON⁽²¹⁾ and is called the 'sheared boundary' method. In its simplest form (Figure 5.9) the tube containing the two solutions is divided with the upper half held in a hole in a disc A over a depression in another disc so that a drop of the solution hangs from the end of the tube. The other half is held in the disc B immediately under a cut or indent in the disc A and is filled with the other solution until a drop protrudes at the end.

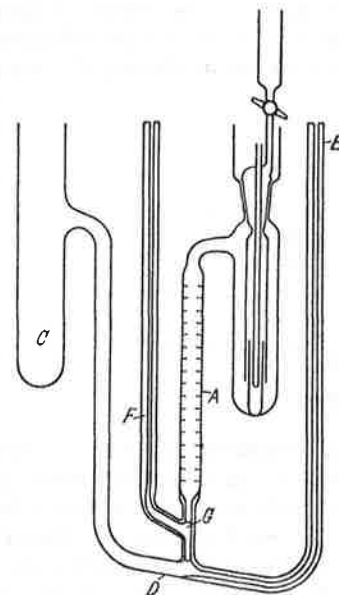


Figure 5.10. From HARTLEY, G. S. and DONALDSON, G. W., *Trans. Faraday Soc.*, 33 (1937) 457

If the two discs have plane surfaces and the discs are moved over one another so that the two tubes are adjacent the excess of each liquid is 'sheared off' and a sharp boundary is formed. The second method is the 'autogenic' method of FRANKLIN and CADY⁽²²⁾ in which the indicator solution is formed by making the anode at the bottom of the tube of a metal such as cadmium, with a solution of potassium chloride over it: on passing a current, a solution of cadmium chloride is formed and a boundary is produced between this solution and the potassium chloride solution. Again, an anode of silver can be used to give a boundary between a silver nitrate solution and a potassium nitrate solution. The third method is the 'air bubble' method⁽²³⁾. The apparatus is shown in Figure 5.10. To the top of the capillary tube F , two pinchcocks are attached, one closed and one half open. At the beginning of the experiment, the whole apparatus

is filled with the leading solution and by closing one pinchcock an air bubble is formed by compressing the air in the capillary *F*; this expands into *A* at *G* separating the solution into two parts. That in *C*, *D* and *E* is removed, these portions of the apparatus are rinsed several times with water and filled with indicator solution. By unscrewing the pinchcock the air bubble can now be withdrawn into *F* and a boundary formed at *G*.

If the boundary moves up the tube, it is necessary to have the indicator solution of greater density than the leading solution and, conversely, if the indicator solution is to be on the top of the leading

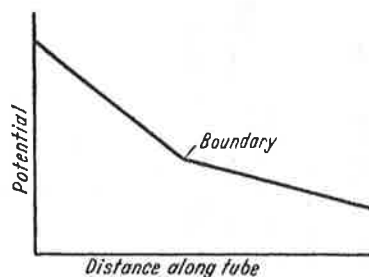


Figure 5.11

solution and the boundary is to move downwards, then the indicator solution should be lighter. Moreover, in whichever direction the boundary moves, the ion of the indicator solution must have a mobility lower than the ion of the leading solution. Fortunately, there is a kind of self-regulating effect which restores the sharpness of the boundary if it is for some reason diminished. Let us consider what will be the concentration of the indicator ion N^+ behind the moving boundary. We have already seen that during the passage of an amount, Q coulombs, of electricity all the M^+ ions between *cd* and *ab* cross the boundary *cd*. Let a further Q coulombs pass: then all the N^+ ions between *ab* and *cd* must cross the plane at *cd* and

$$t'_1 = \frac{Vc_{N^+}F}{Q}$$

where t'_1 is the transport number of the N^+ ion at the concentration c_{N^+} at which it is present in the volume V behind the boundary.

Then

$$t_1/t'_1 = c_{M^+}/c_{N^+}$$

This is sometimes called Kohlrausch's regulating function. Now t'_1 must be less than t_1 and therefore c_{N^+} must be less than c_{M^+} . Thus if we consider the fall in potential along the tube (Figure 5.11),

both the lower mobility and the lower concentration of the indicator ion will combine to produce a sharper drop in potential in the indicator solution than in the leading solution. If by any mischance a leading ion diffuses into the indicator solution its comparatively high mobility shoots it forward again down the comparatively large potential gradient; conversely should an indicator ion find itself too far ahead, its lower mobility combined with the lower gradient slows it up: in either event, there is a mechanism at work to restore the sharpness of the boundary. The same mechanism adjusts the concentration c_{N^+} of the indicator solution: if c_{N^+} is initially too large, the potential gradient is lower than that demanded by the condition $t_1/t'_1 = c_{M^+}/c_{N^+}$ and the N^+ ion travels more slowly until the correct value of c_{N^+} is reached, and conversely, if the initial value of c_{N^+} is too small. Naturally it cannot be expected that this 'self-regulating mechanism' can cope with wide departures from ideal conditions. MacInnes has found by experiment, however, that considerable tolerance is permitted so that it is sufficient to adjust the concentration of the indicator solution to within only 5 to 10 per cent of the required value.

During the passage of the current there may be volume changes resulting from electrode reactions, etc. Except in concentrated solutions, the correction for this effect is small but it may be considered here in some detail because of its bearing on the distinction between the Hittorf and the moving boundary transport numbers, a matter appreciated very clearly by Lewis⁽²⁴⁾ as long ago as 1910. The moving boundary method gives an ionic mobility relative to the fixed glass tube in which the measurements are made. The Hittorf experiment measures the number of ions crossing a plane fixed relative to a hypothetical plane in the solvent which of course moves if the solvent moves. MACINNES and LONGWORTH⁽²⁵⁾ illustrate this by reference to an experiment (Figure 5.12) in which potassium chloride forms the leading solution, barium chloride the indicator solution and the tube is sealed at the bottom by a silver electrode; x marks the position of a water molecule of a hypothetical type, hypothetical since it is presumed free of Brownian motion. *cd* is a plane in a region not subject to change of concentration. On passing a faraday of electricity the following changes occur:

1. t_1 equivalents of K^+ cross the plane *cd* in an upward direction giving a volume decrease $t_1 V_{K^+}^{KCl}$, i.e., t_1 times the partial molal volume of the potassium ion in potassium chloride solution.
2. t_2 equivalents of Cl^- cross *cd* in a downward direction with a volume increase $t_2 V_{Cl^-}^{KCl}$.